RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution)

Rajalakshmi Nagar, Thandalam – 602 105

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

CR23532 – CYBER AND DIGITAL FORENSICS

LABORATORY RECORD

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EXPERIMENT – 1

Study of Computer Forensics and Different Tools Used for Forensic Investigation

Aim

To examine the discipline of computer (digital) forensics and to become familiar with the various tools that support a forensic investigation.

Introduction

Digital forensics is the branch of forensic science that is concerned with identifying the person responsible for a computer crime or a digital intrusion. Investigators apply a broad set of techniques to attribute an incident to its perpetrator. The discipline rests on a simple principle: anyone who commits an offence on a digital system unknowingly leaves traces of themselves behind. These traces may take the form of log entries, registry modifications, attack tools, malware or fragments of deleted files, and each of them can act as a clue or as evidence that points towards the identity of the offender.

As with hacking, digital forensics is supported by a wide variety of software. Most professional investigators depend on three leading commercial suites – EnCase Forensic (Guidance Software), Forensic Toolkit or FTK (AccessData) and ProDiscover. These suites are costly and are used mainly by law-enforcement agencies. Comparable capabilities are offered by free and open-source packages such as The Sleuth Kit (TSK), Helix and Knoppix, although they lack the refined interface of the commercial products. Many standalone utilities are also bundled with security distributions such as BackTrack and Kali Linux.

Procedure

- The commercial forensic suites relied upon by law-enforcement agencies – EnCase Forensic, Forensic Toolkit (FTK) and ProDiscover – were identified and their main features were noted.
- The free and open-source alternatives – The Sleuth Kit (TSK), Helix and Knoppix – were examined.
- The standalone forensic utilities shipped with the Kali Linux / BackTrack distribution were listed along with the purpose of each.

- Field tools used in practical investigations, namely FTK Imager, Log Parser Lizard GUI, Noxcivis Field Toolkit, Active@ Partition Recovery and Autopsy, were studied.

The role of every tool covered in this study is summarised in the table below.

Forensic Tools Studied

Tool	Purpose
EnCase Forensic	Commercial investigation suite commonly relied on by police and law-enforcement agencies.
FTK (Forensic Toolkit)	Commercial suite from AccessData; its FTK Imager component produces forensically sound disk images.
ProDiscover	Commercial suite for disk investigation and evidence analysis.
The Sleuth Kit (TSK)	Open-source set of command-line tools for file-system analysis.
Autopsy	Graphical interface built on The Sleuth Kit for examining disks and disk images.
FTK Imager	Creates disk images, captures memory and recovers files.
Log Parser Lizard GUI	Parses and queries log files of many different formats.
Noxcivis Field Toolkit	Gathers data from a live system during incident response.
Active@ Partition Recovery	Restores deleted or lost partitions.
USBDeview	Shows USB devices that are connected now or were connected earlier.
LastActivityView	Presents a timeline of recent user actions and system events.
ChromeHistoryView / IECacheView	Read browser history and cached files.
Compare It	Compares two files and marks the differences between them.
DumpIt	Captures the physical memory (RAM) of a Windows machine.
Bulk Extractor	Pulls out features such as e-mail addresses, URLs and phone numbers from an image.
ExifReader	Reads the EXIF metadata stored in image files.
Recover My Files	Restores deleted files from drives and disk images.

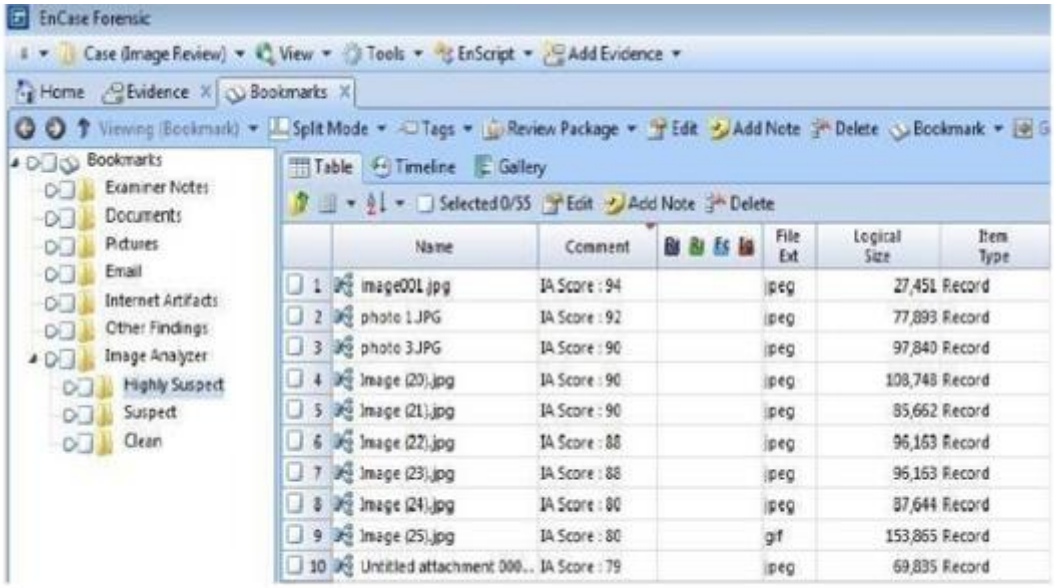


Fig. 1.1 EnCase Forensic – bookmarked images classified by the Image Analyzer

Result

Hence, the discipline of computer forensics was studied, and the purpose and usage of the major tools employed in a forensic investigation were understood successfully.

EXPERIMENT – 2

Introduction to Digital Forensics (TryHackMe)

Aim

To gain a foundational understanding of digital forensics and of the standard investigation process by completing the TryHackMe room "Intro to Digital Forensics".

Procedure

- Sign in to TryHackMe (tryhackme.com) and open the room titled "Intro to Digital Forensics".
- Go through Task 1 – Introduction to Digital Forensics – which explains what digital forensics is and where it is applied, such as criminal cases, corporate incident response and civil disputes.
- Go through Task 2 – Digital Forensics Process – and record the standard stages of an investigation: identification, preservation, collection, examination, analysis and presentation (reporting).
- Complete Task 3 – Practical Example of Digital Forensics – in which a sample scenario is examined to trace the origin of the evidence.
- Submit the answers to the questions in each task, referring to the hints given in the room wherever required.
- Verify that the room shows 100% completion and summarise the key learnings.

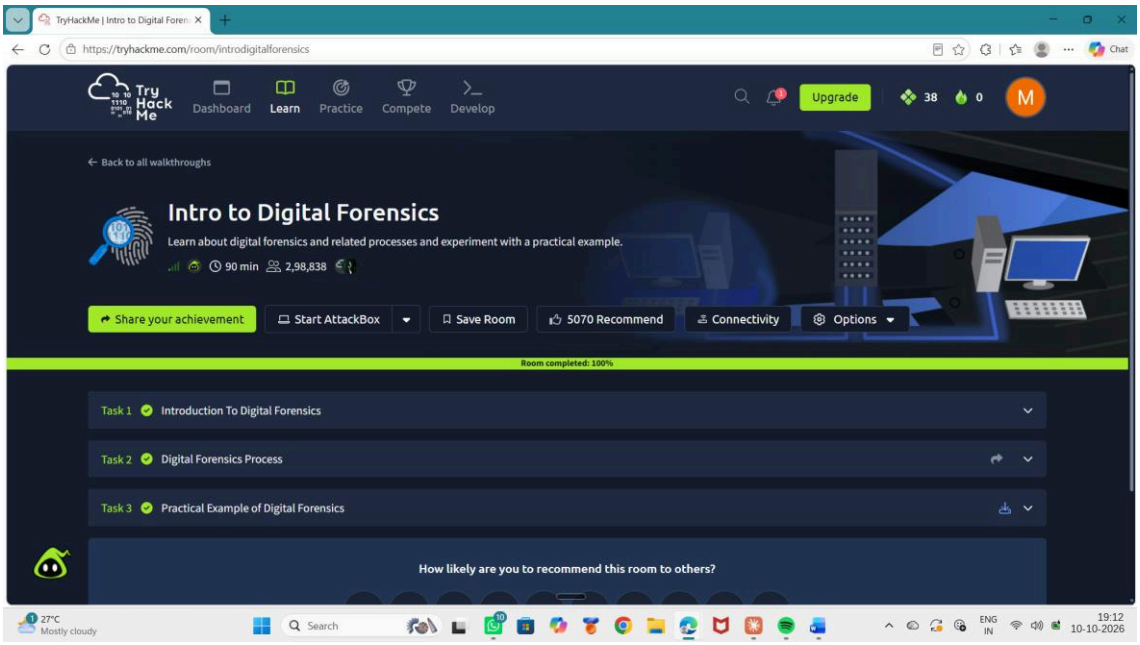


Fig. 2.1 TryHackMe – "Intro to Digital Forensics" room completed (100%)
TIMESTAMP: 10-10-2026, 19:12

Result

The TryHackMe room "Intro to Digital Forensics" was completed with 100% progress, and the fundamentals of digital forensics along with the standard investigation process were understood.

EXPERIMENT – 3

Windows Forensics 1 (TryHackMe)

Aim

To study Windows registry forensics and to recognise the key forensic artifacts maintained by Windows by completing the TryHackMe room "Windows Forensics 1".

Procedure

- Sign in to TryHackMe and launch the room "Windows Forensics 1".
- Read "Introduction to Windows Forensics" to understand why Windows systems hold a large amount of evidence and which artifacts they maintain.
- Read "Windows Registry and Forensics" and learn the purpose of each registry hive – SAM, SYSTEM, SOFTWARE, SECURITY, NTUSER.DAT and USRCLASS.DAT.
- Read "Accessing Registry Hives Offline" to see how the hives taken from an acquired image are loaded and examined with tools such as Registry Explorer.
- Go through "Data Acquisition" to learn how the required hives are collected from a live or a powered-off system.
- Work through "Exploring Windows Registry" and "System Information and System Accounts" – computer name, last shutdown time, user accounts and login details.
- Work through "Usage or Knowledge of Files/Folders" and "Evidence of Execution" – recently used files, shellbags, Run keys, prefetch files and jump lists.
- Study "External Devices / USB Device Forensics", which covers the registry entries left behind when USB devices are connected.
- Attempt the hands-on challenge and go through the concluding task of the room.

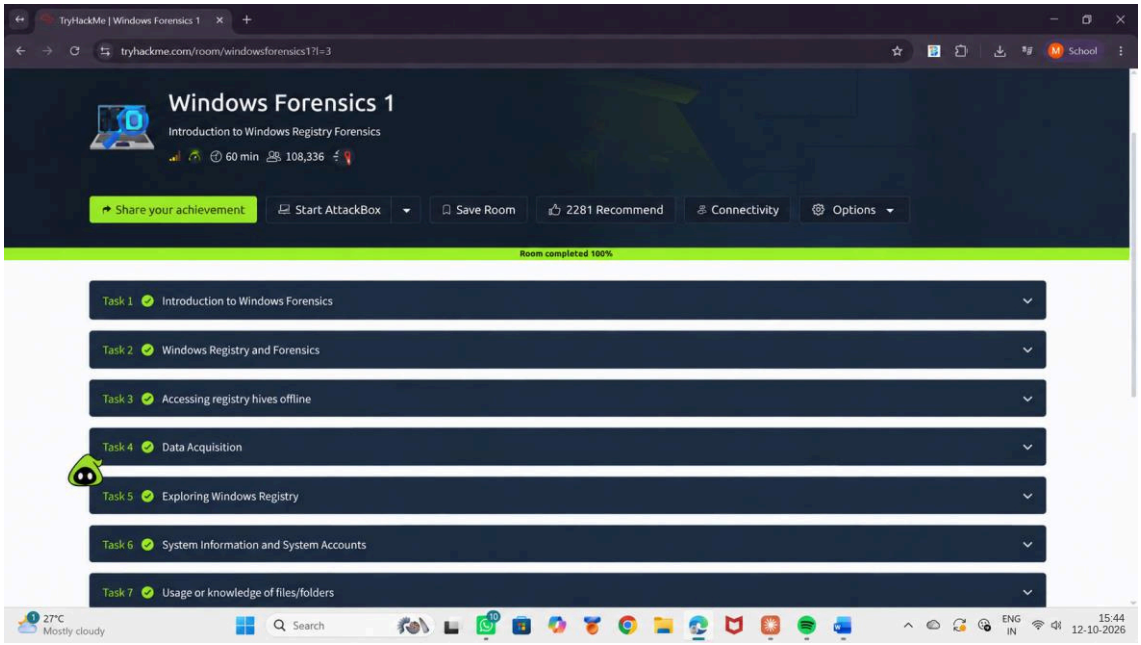


Fig. 3.1 TryHackMe – "Windows Forensics 1" room with all tasks completed
TIMESTAMP: 12-10-2026, 15:44

Result

The room "Windows Forensics 1" was completed successfully, and the registry hives, evidence-of-execution artifacts, file and folder usage traces and USB device forensics were understood.

EXPERIMENT – 4

Live Forensics Case Investigation using Autopsy

Aim

To carry out a forensic case investigation on a disk image with the Autopsy platform and to produce a forensic report of the findings.

Procedure

- Download Autopsy from its official website, install it and start the application; the welcome window offers the 'New Case' and 'Open Case' options.
- Choose 'New Case', give the case a name and select a base directory in which the case database and the reports will be saved.
- On the Optional Information page, enter the case number and the examiner details (name, phone number, e-mail address and notes), then click 'Finish' to create the case.
- The Add Data Source wizard appears. Select the host in the first step and pick the data source type in the second step (Disk Image or VM File, Local Disk, Logical Files, Unallocated Space Image File, etc.).
- Browse to the image file that has to be analysed (here D:\Cases\Drive_Evidence_01.dd), set the input time zone to Asia/Calcutta and choose whether the image should be broken into chunks.
- In the Configure Ingest step, tick the modules to be run – Hash Lookup, File Type Identification, Extension Mismatch Detector, Embedded File Extractor, Picture Analyzer, Email Parser, Encryption Detection, PhotoRec Carver, Data Source Integrity, EXIF Parser, etc. – so that as much evidence as possible is extracted.
- Click 'Finish'; Autopsy analyses the data source and stores the findings in the case database.
- After the ingest completes, open 'Generate Report', select the HTML Report module, enter a header and a footer, and generate the report.
- Review the report and the results tree – ingest history, enabled modules, encryption detected, operating system information, user accounts, recent documents, web history, web downloads, cookies and e-mail addresses.

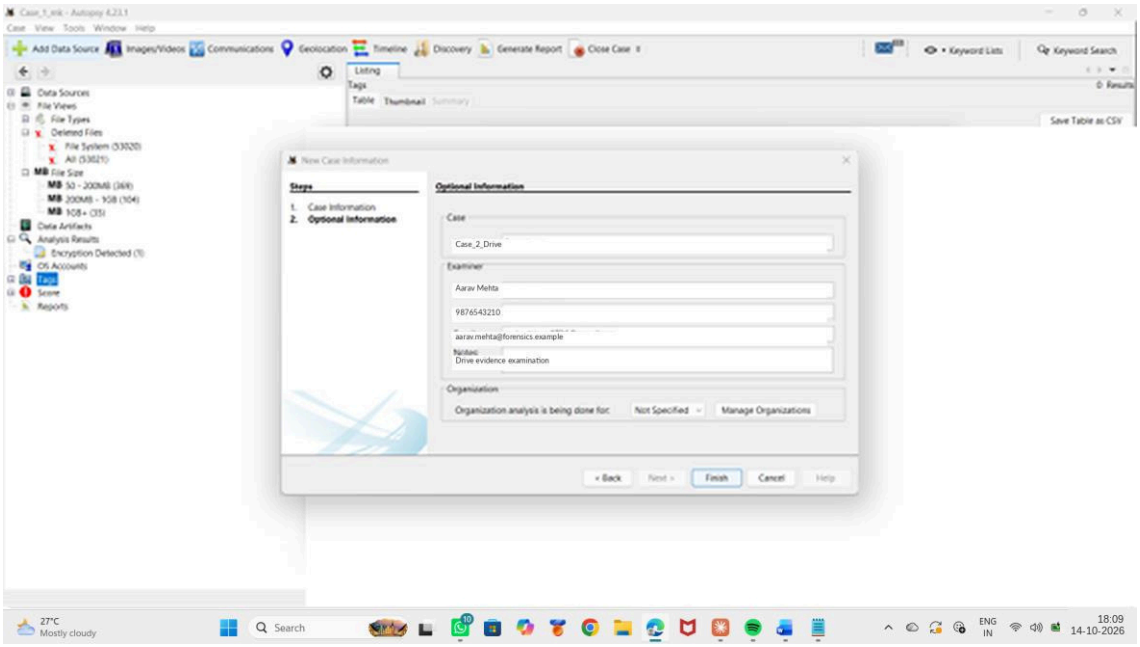


Fig. 4.1 Autopsy – entering the case and examiner information

TIMESTAMP: 14-10-2026, 18:09

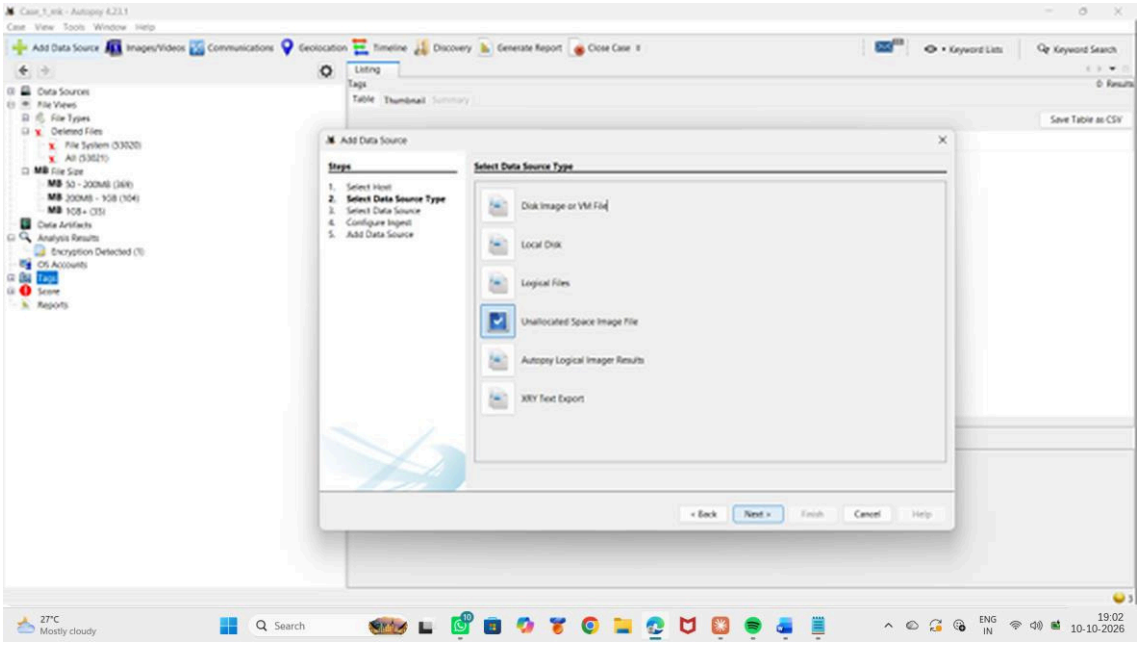


Fig. 4.2 Autopsy – Add Data Source wizard: choosing the data source type

TIMESTAMP: 10-10-2026, 19:02

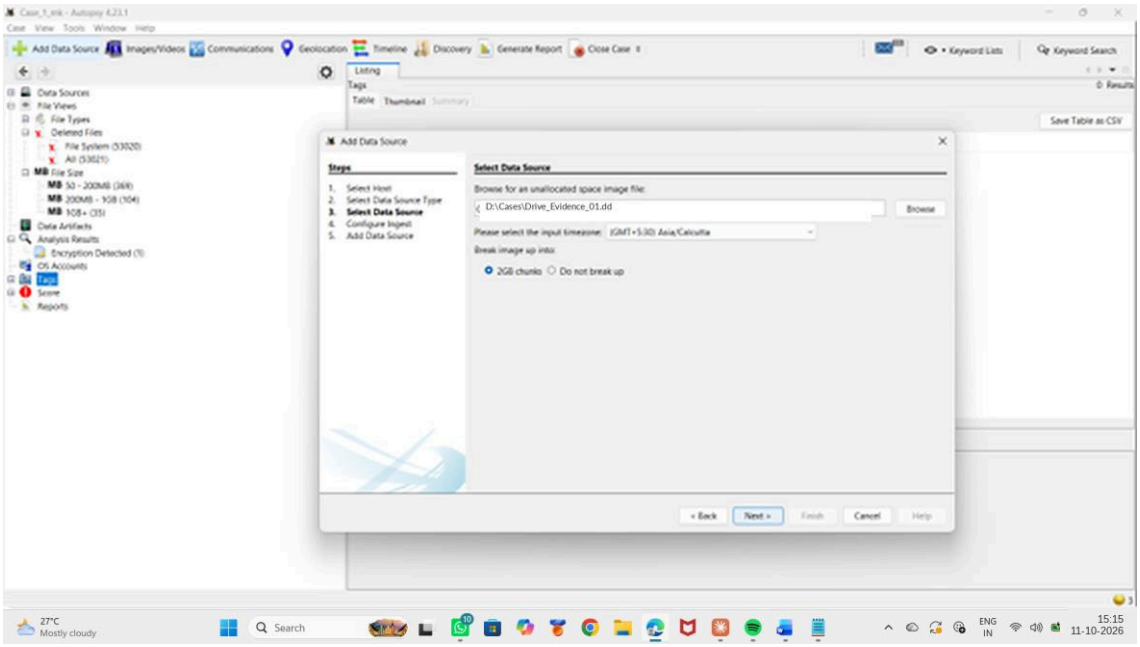


Fig. 4.3 Autopsy – browsing for the image file and setting the input time zone

TIMESTAMP: 11-10-2026, 15:15

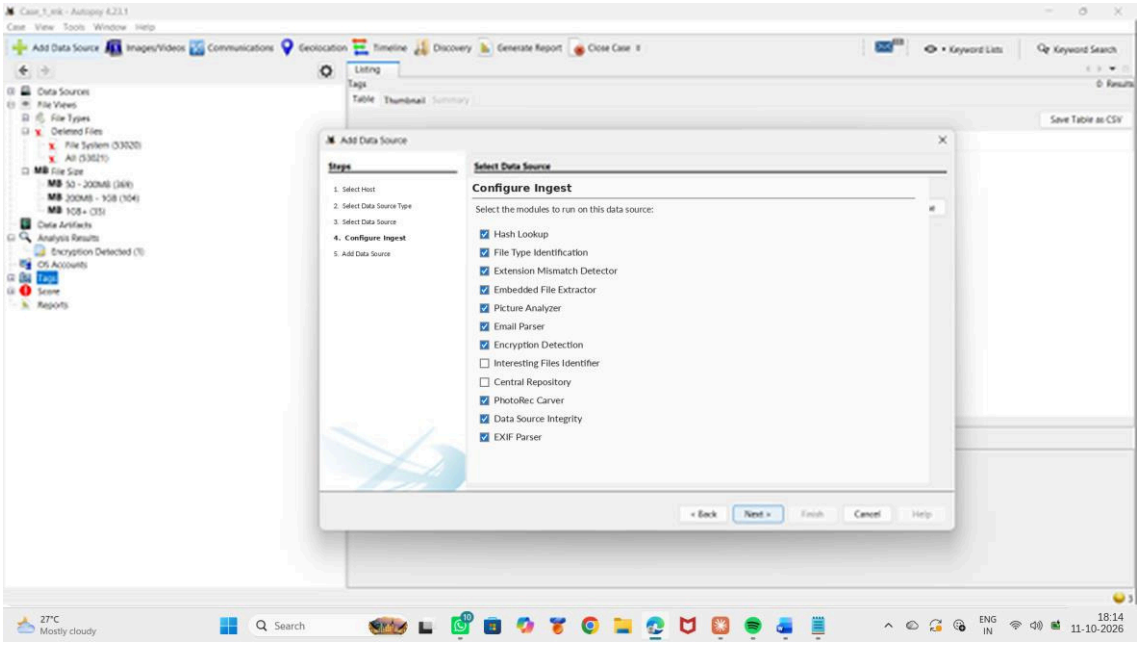


Fig. 4.4 Autopsy – selecting the ingest modules to run on the data source

TIMESTAMP: 11-10-2026, 18:14

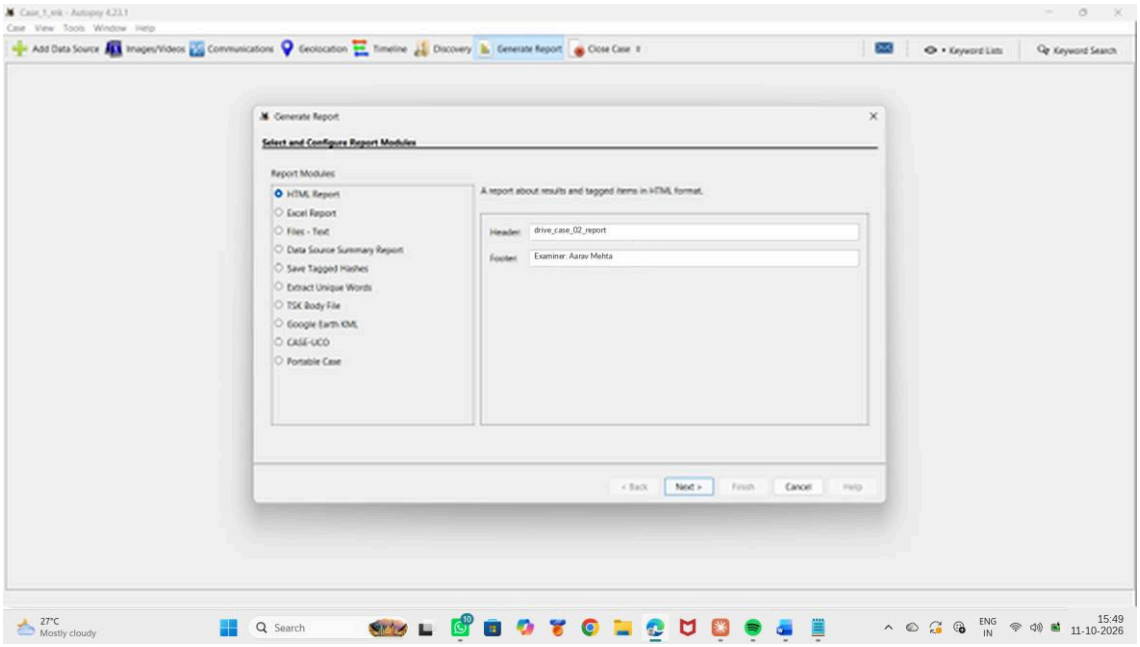


Fig. 4.5 Autopsy – Generate Report: selecting and configuring the HTML report module

TIMESTAMP: 11-10-2026, 15:49

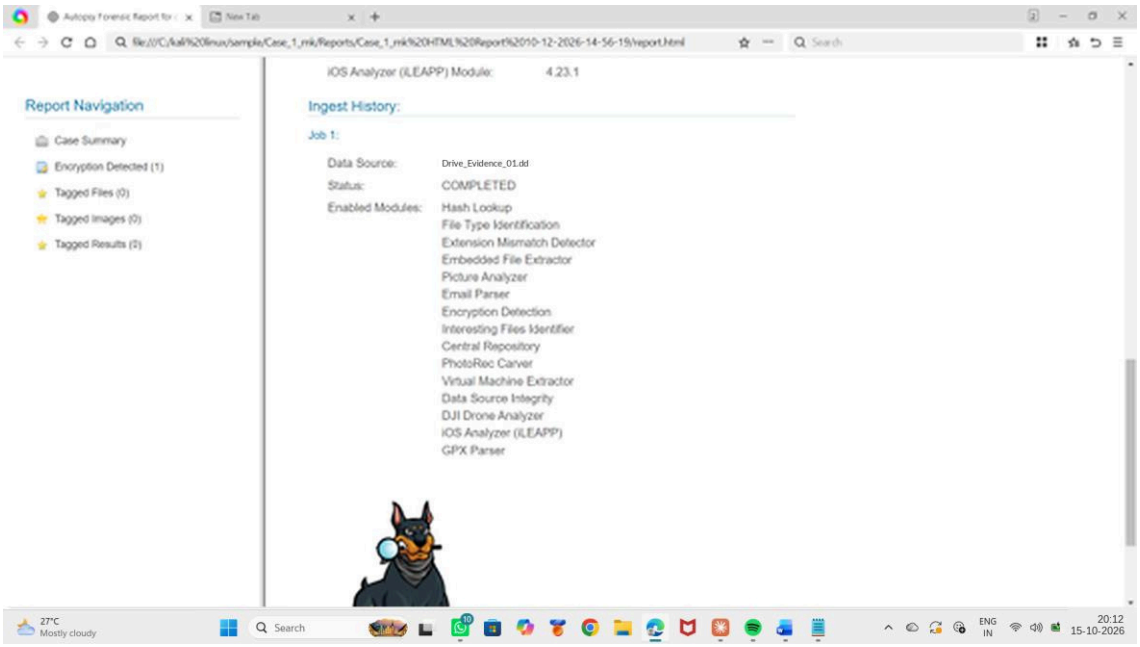


Fig. 4.6 Autopsy forensic report – ingest history showing the completed job and enabled modules

TIMESTAMP: 15-10-2026, 20:12

Result

Hence, a forensic case investigation was performed on a disk image using Autopsy. The data source was ingested with the selected modules, the evidence artifacts were examined and an HTML forensic report was generated successfully.

EXPERIMENT – 5

Recover Deleted Files using Forensics Tools (FTK Imager)

Aim

To recover a deleted file from storage media by acquiring a forensically sound bit-stream image with FTK Imager, and to confirm the integrity of that image through hash verification.

Procedure

- **Step 1 (Create the file):** Open Notepad, type a short message (here "attack is gonna happen . save the system") and save it as attack.txt.
- **Step 2 (Delete the file):** Delete the file so that it is no longer visible to the user. Its contents remain in the unallocated area of the disk until they are overwritten by new data.
- **Step 3 (Need for imaging):** An ordinary copy or backup does not include deleted data, so the investigator first acquires a bit-by-bit forensic image of the media.
- **Step 4 (Launch the imager):** Install and run FTK Imager and choose File → Create Disk Image.
- **Step 5 (Select the source):** In the Select Source window, pick the evidence type – Physical Drive, Logical Drive, Image File or Contents of a Folder – and click 'Next'.
- **Step 6 (Image type):** Select the output format – Raw (dd), SMART, E01 or AFF. The E01 format is preferred because it supports compression and built-in integrity checks.
- **Step 7 (Evidence details):** Fill in the Evidence Item Information – case number, evidence number, a unique description, the examiner name and notes.
- **Step 8 (Destination):** Provide the image destination folder, the image file name, the fragment size (MB) and the compression level; AD Encryption can be enabled to protect the image with a password.
- **Step 9 (Create the image):** Click 'Start'; FTK Imager builds the bit-by-bit image and displays the progress, the elapsed time and the estimated time remaining.
- **Step 10 (Verify the image):** Once imaging is finished, run File → Verify Drive/Image to compute the MD5 and SHA-1 values and compare them with the reported hashes, proving that the image is an exact copy of the source.
- **Step 11 (Recover the file):** Analyse the verified image to carve out and recover the deleted file, and compare the recovered contents with the original to confirm the recovery.

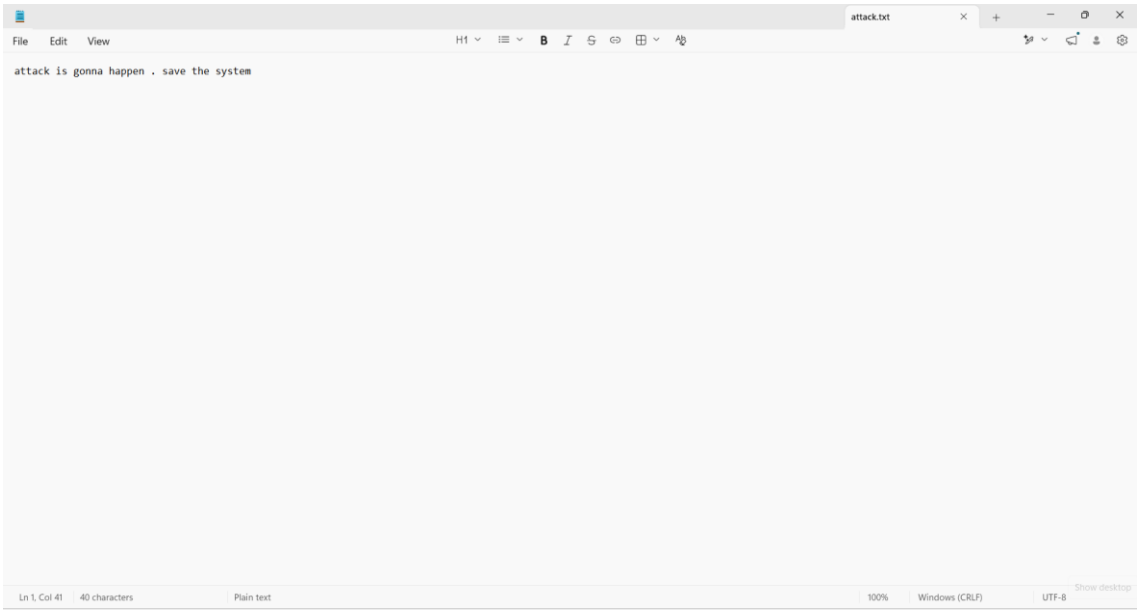


Fig. 5.1 Creating the target file attack.txt with the sample message
TIMESTAMP: 10-10-2026, 15:11

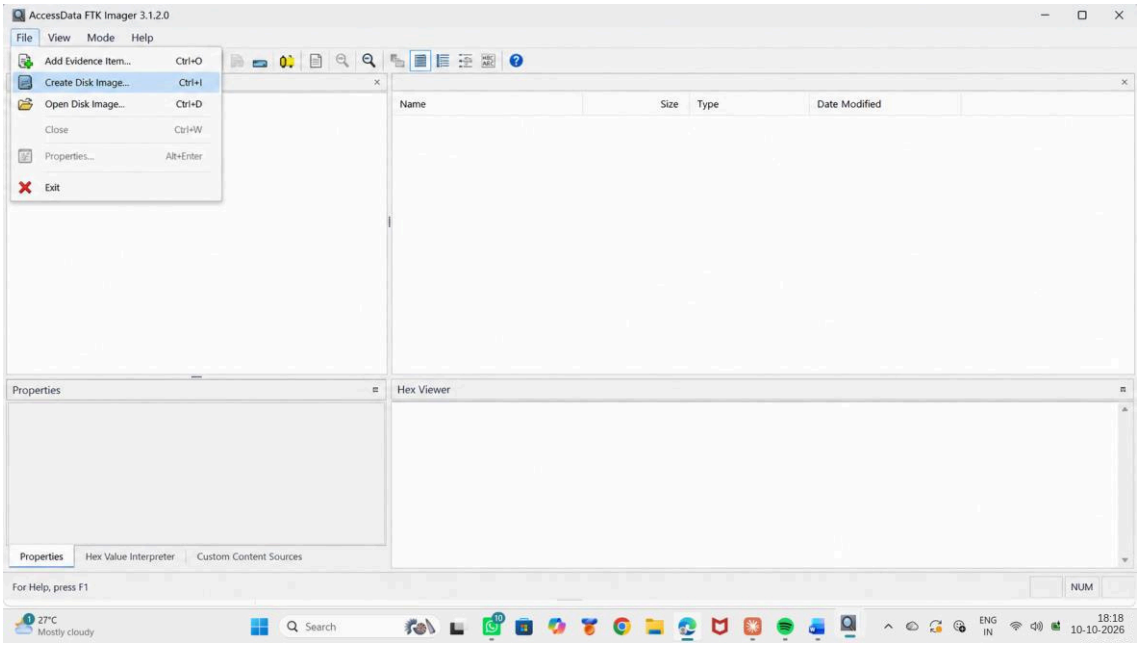


Fig. 5.2 FTK Imager – choosing 'Create Disk Image' from the File menu
TIMESTAMP: 10-10-2026, 18:18

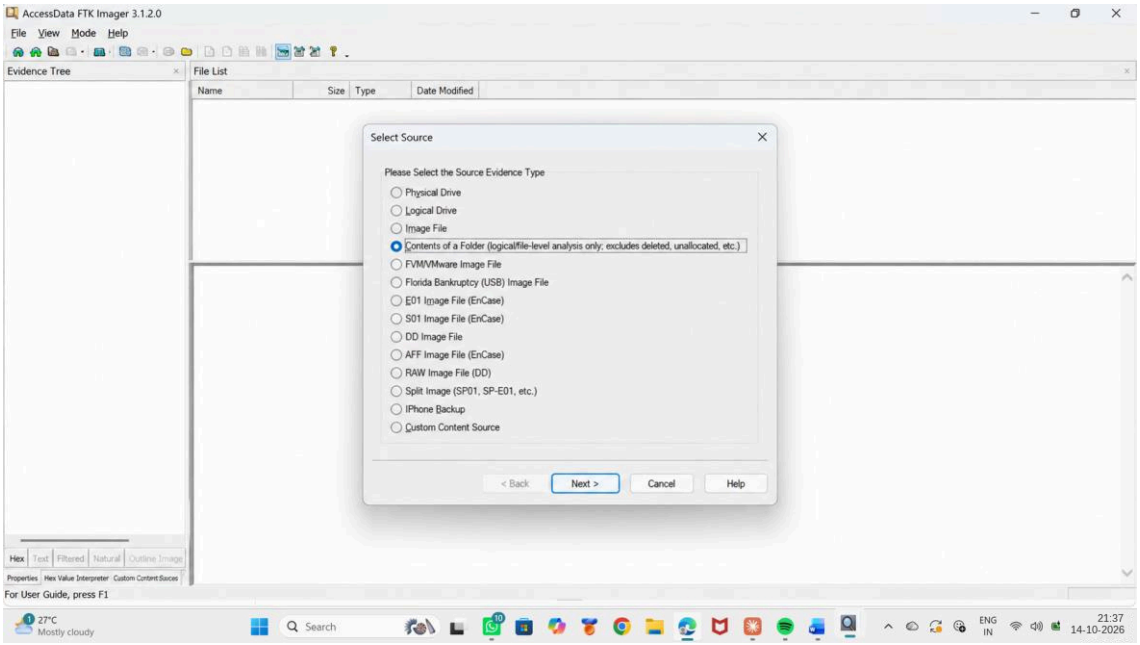


Fig. 5.3 FTK Imager – selecting the source evidence type

TIMESTAMP: 14-10-2026, 21:37

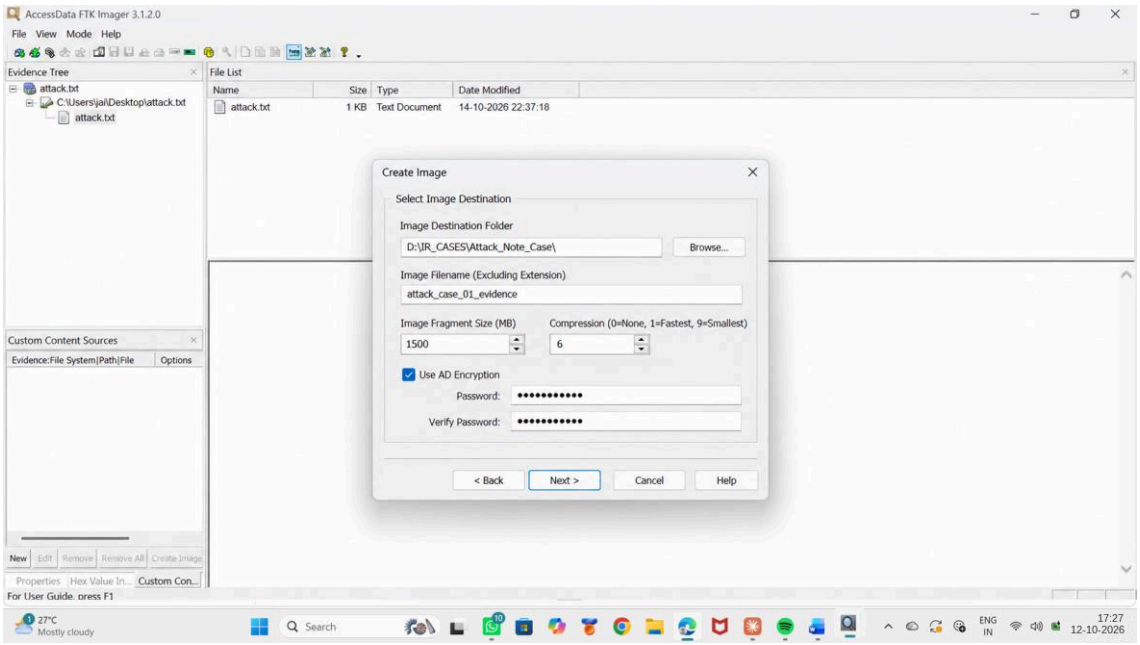


Fig. 5.4 FTK Imager – specifying the image destination folder and file name

TIMESTAMP: 12-10-2026, 17:27

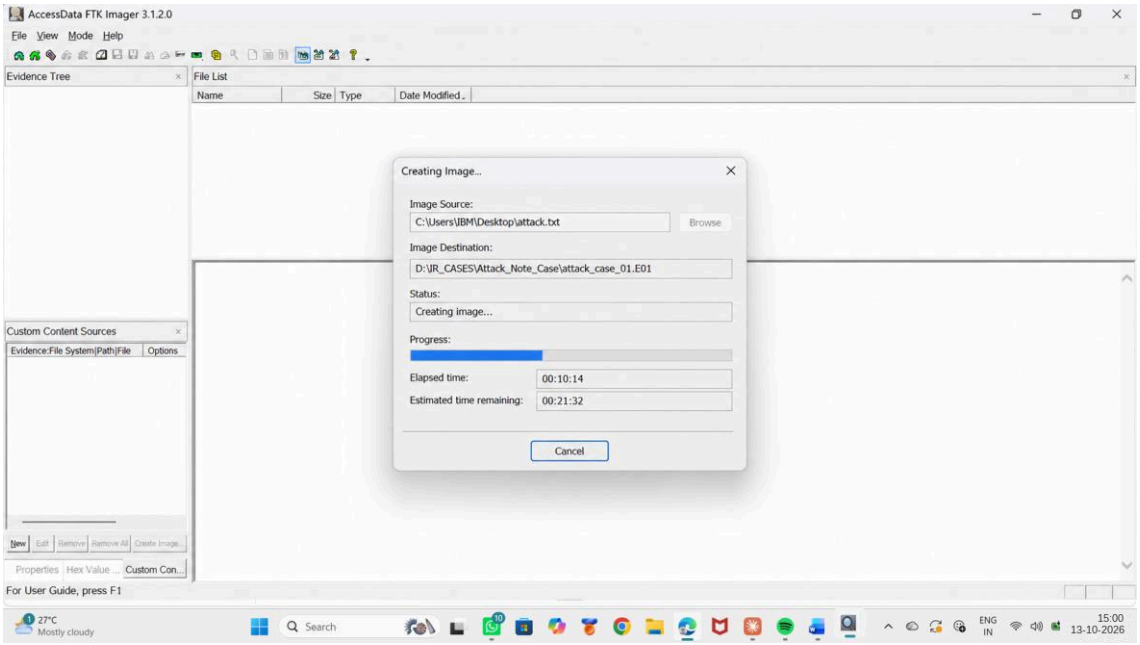


Fig. 5.5 FTK Imager – creating the image attack_case_01.E01

TIMESTAMP: 13-10-2026, 15:00

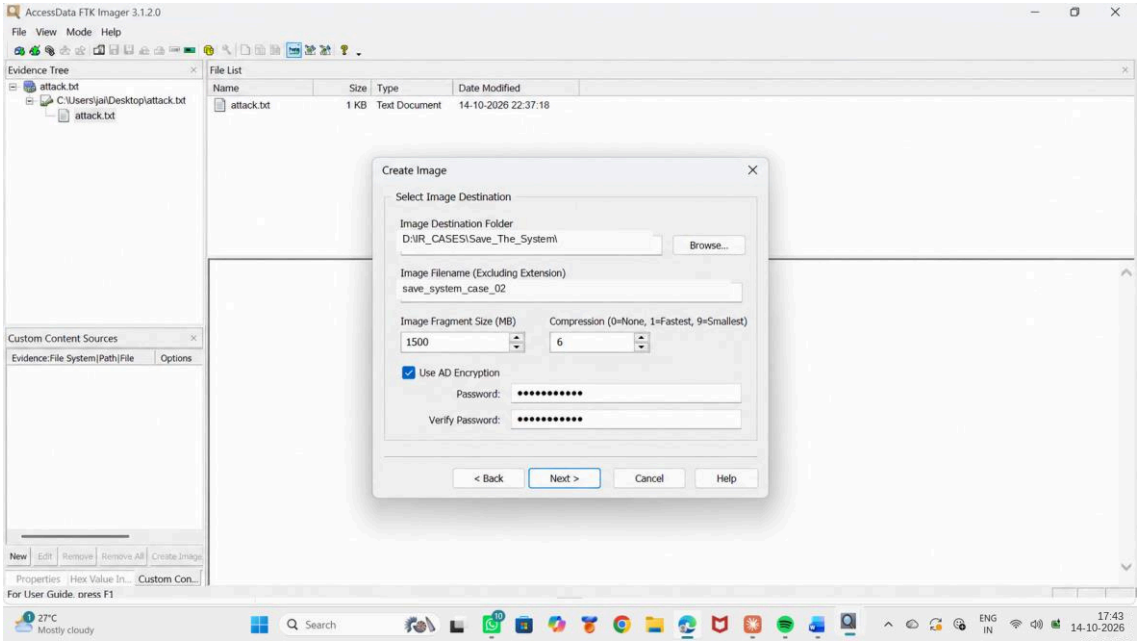


Fig. 5.6 FTK Imager – setting the destination for the second image (save_system_case_02)

TIMESTAMP: 14-10-2026, 17:43

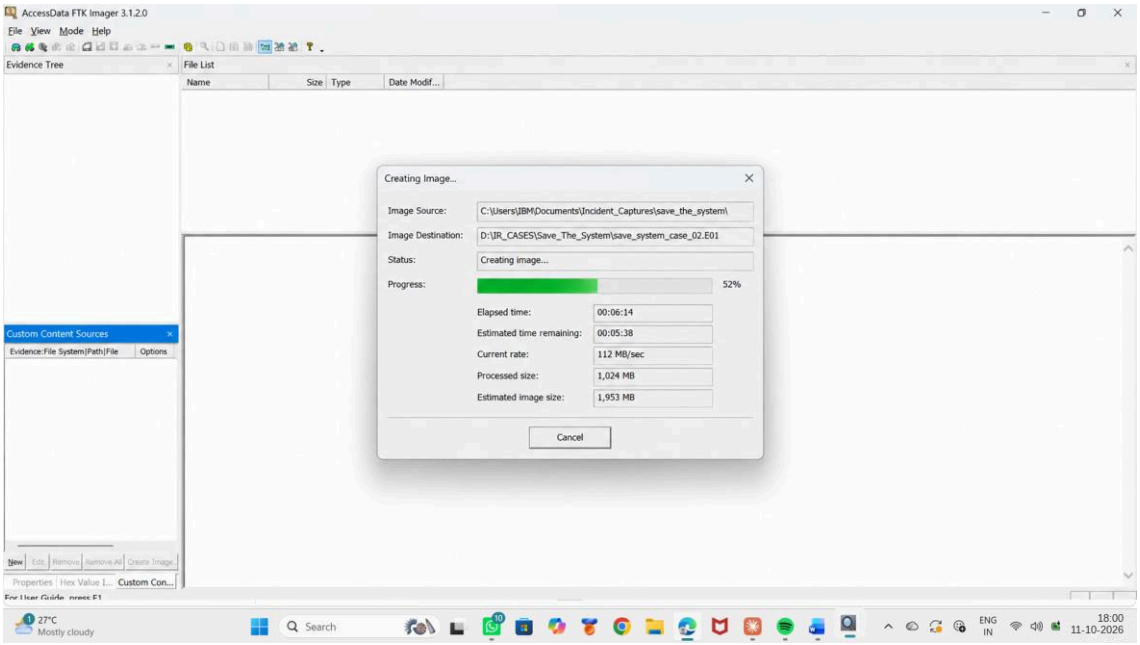


Fig. 5.7 FTK Imager – imaging of save_system_case_02.E01 in progress
TIMESTAMP: 11-10-2026, 18:00

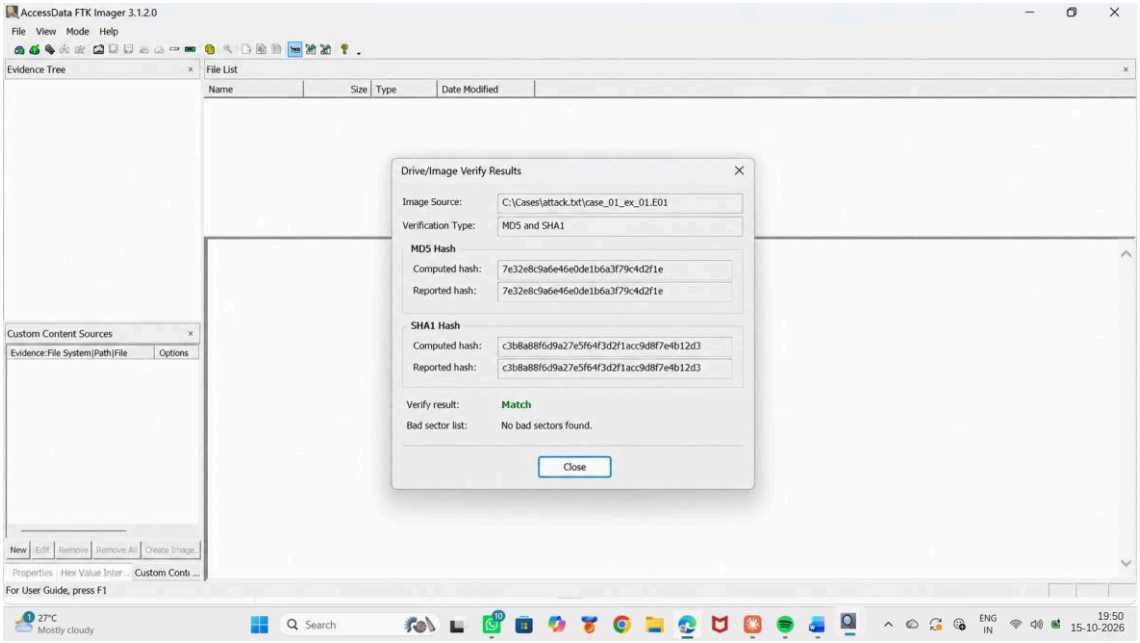


Fig. 5.8 FTK Imager – Drive/Image Verify Results with matching MD5 and SHA-1 hashes
TIMESTAMP: 15-10-2026, 19:50

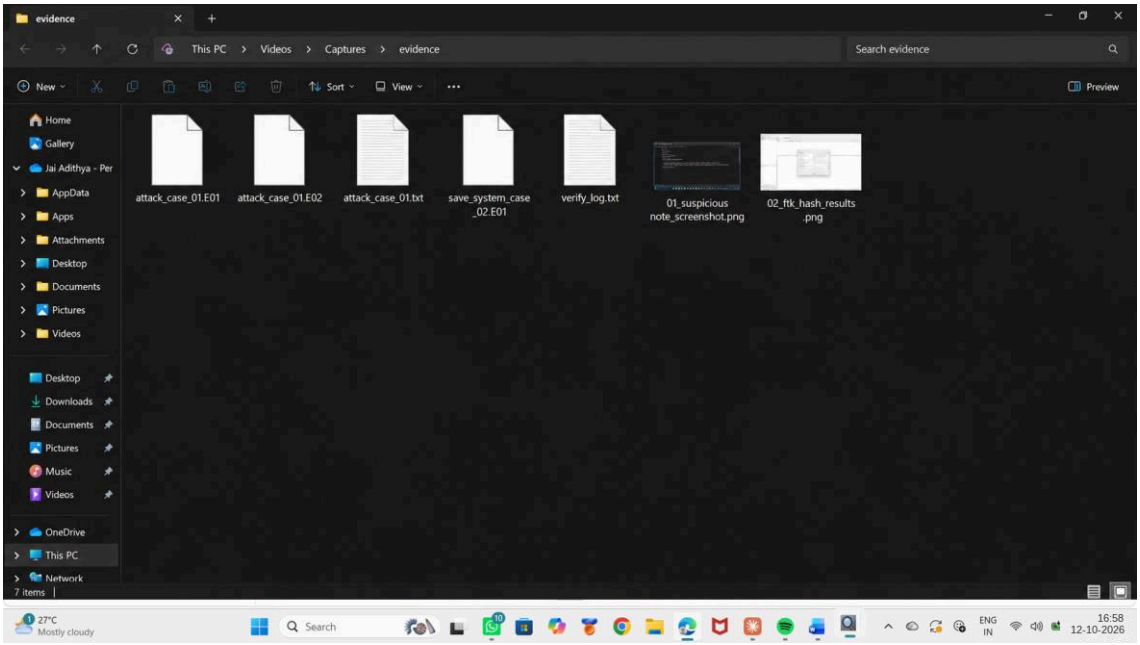


Fig. 5.9 The generated E01 image segments and verification files in the evidence folder TIMESTAMP: 12-10-2026, 16:58

Result

Hence, the deleted file was recovered with the help of forensic tools. A bit-by-bit E01 image of the evidence was acquired using FTK Imager, its integrity was confirmed through matching MD5 and SHA-1 hash values, and the deleted content was recovered and verified successfully.

EXPERIMENT – 6

Find Last Connected USB on the System (USB Forensics)

Aim

To determine the USB device that was most recently connected to the system with the help of USBDeview.

Procedure

- Download USBDeview from NirSoft and run the executable on the system under examination.
- The utility lists every USB device that is currently plugged into the computer, together with the devices that were connected to it earlier.
- For each entry, note the extended details – device name and description, device type, serial number (for storage devices), Registry Time 1 and Registry Time 2 (when the device was installed / last connected), VendorID and ProductID.
- Plug in the USB flash drive and locate its record in the list, for example the 'SanDisk Ultra USB Device' of type Mass Storage.
- Double-click the record to open its Properties and confirm the serial number, drive letter, computer name and the install and connect times.
- Note down these details as evidence. USBDeview can additionally uninstall, enable, disable or disconnect USB devices, and it can also be run against a remote computer.

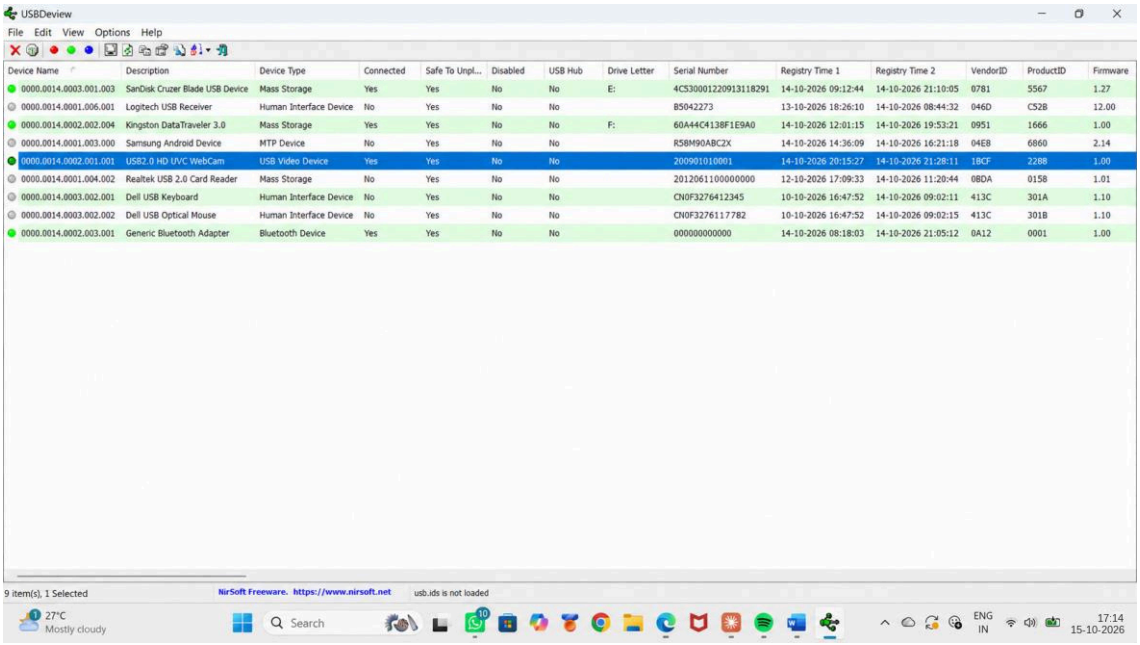


Fig. 6.1 USBDeview – currently connected and previously used USB devices

TIMESTAMP: 15-10-2026, 17:14

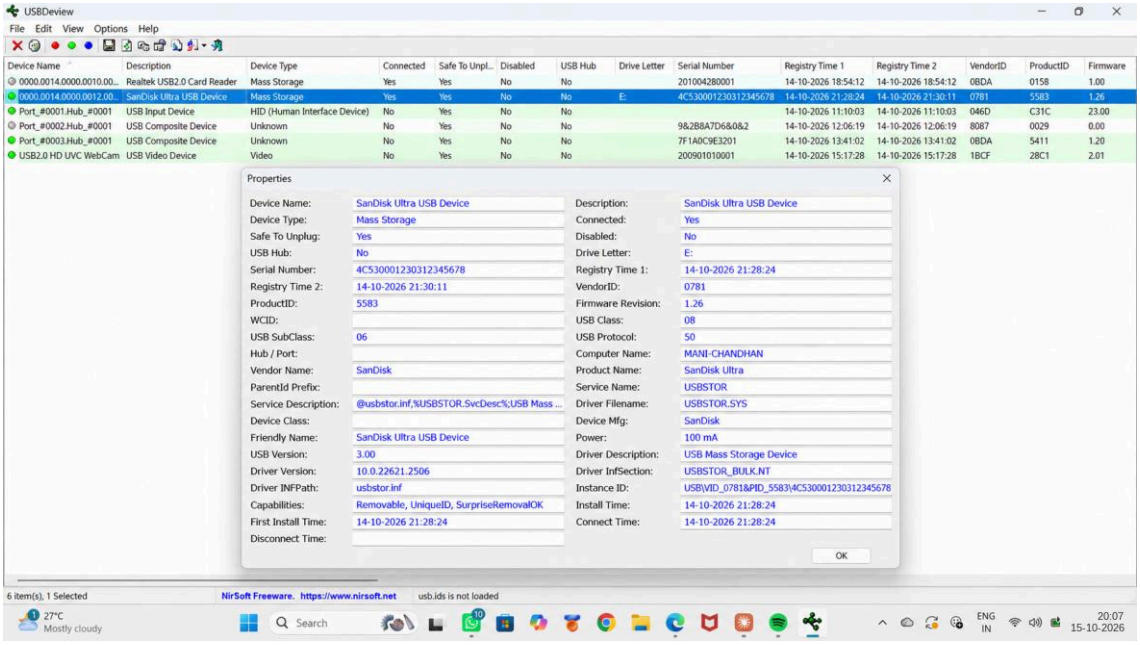


Fig. 6.2 USBDeview – properties of the last connected USB mass-storage device

TIMESTAMP: 15-10-2026, 20:07

Result

Hence, the most recently connected USB device was identified using USBDeview, and its device name, serial number, VendorID/ProductID, drive letter and connection time were recorded as evidence.

EXPERIMENT – 7

View Last Activity of Your PC

Aim

To examine the recent activity carried out on the system using the LastActivityView utility.

Procedure

- Download LastActivityView (NirSoft) and launch it.
- The tool gathers information from several sources on the running system and presents a log of the actions performed by the user and the events that occurred on the computer.
- Observe the columns that are displayed – Action Time, Description, Filename, Full Path, More Information, File Extension and Data Source.
- Go through the list to find the latest programs that were executed and the files or folders that were opened (for example chrome.exe, EXCEL.EXE, WINWORD.EXE and Project_Budget.xlsx).
- Right-click a record and choose 'Properties' to see the full details of that activity, including the exact action time and its data source (Prefetch file, registry key, etc.).
- Use the resulting timeline of user and system activity for forensic analysis.

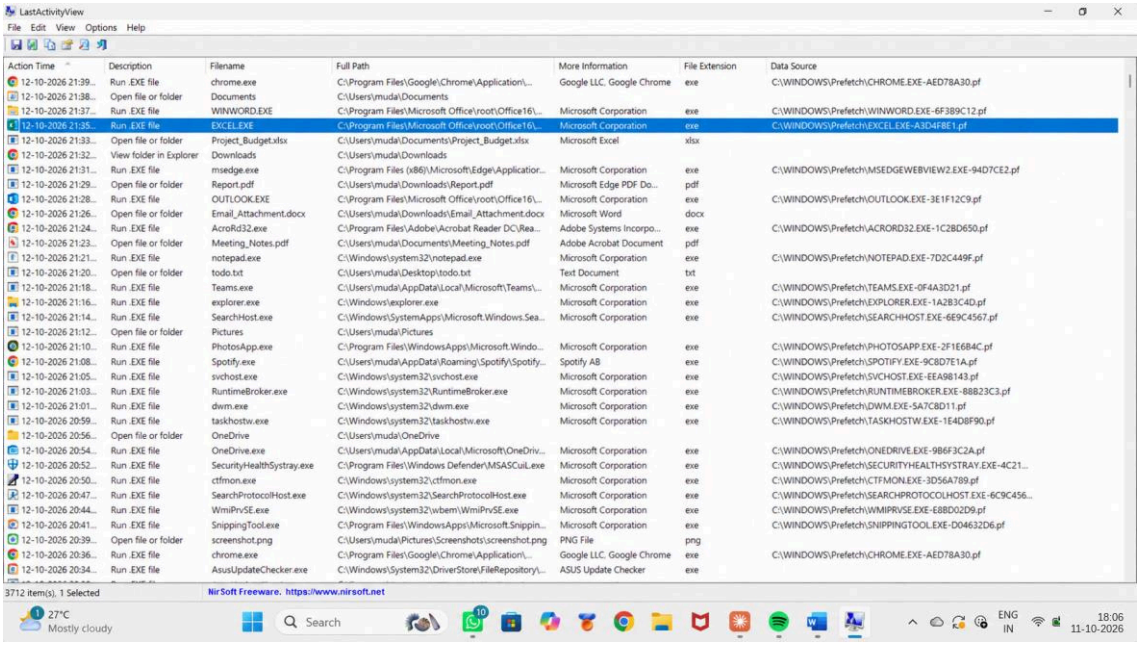


Fig. 7.1 LastActivityView – timeline of recent user actions and system events

TIMESTAMP: 11-10-2026, 18:06

Result

Hence, the recent activities performed on the computer were viewed successfully with LastActivityView, giving a timeline of user actions and system events that supports forensic analysis.

EXPERIMENT – 8

Extraction of Browser Artifacts

Aim

To extract browser artifacts such as the browsing history and cached content from the system using browser forensic utilities.

Procedure

- ChromeHistoryView is a lightweight NirSoft utility that reads the history database of the Google Chrome browser and lists the web pages visited over the recent days.
- For every visited page it shows the URL, page title, visit date and time, visit count, typed count, referrer, visit duration and Visit ID.
- Launch ChromeHistoryView and study the list of visited pages.
- Use the 'Find' option to search the history for a particular keyword (for example 'whatsapp' or 'youtube').
- Open the properties of a record to view details such as the visit count, the referrer and the Chrome profile it belongs to.
- ChromeCacheView reads the cache folder of Chrome and lists the cached files with their URL, content type, size and last-accessed time.
- In the same manner, IEHistoryView and IECacheView can be used to read the history and cache of Internet Explorer.

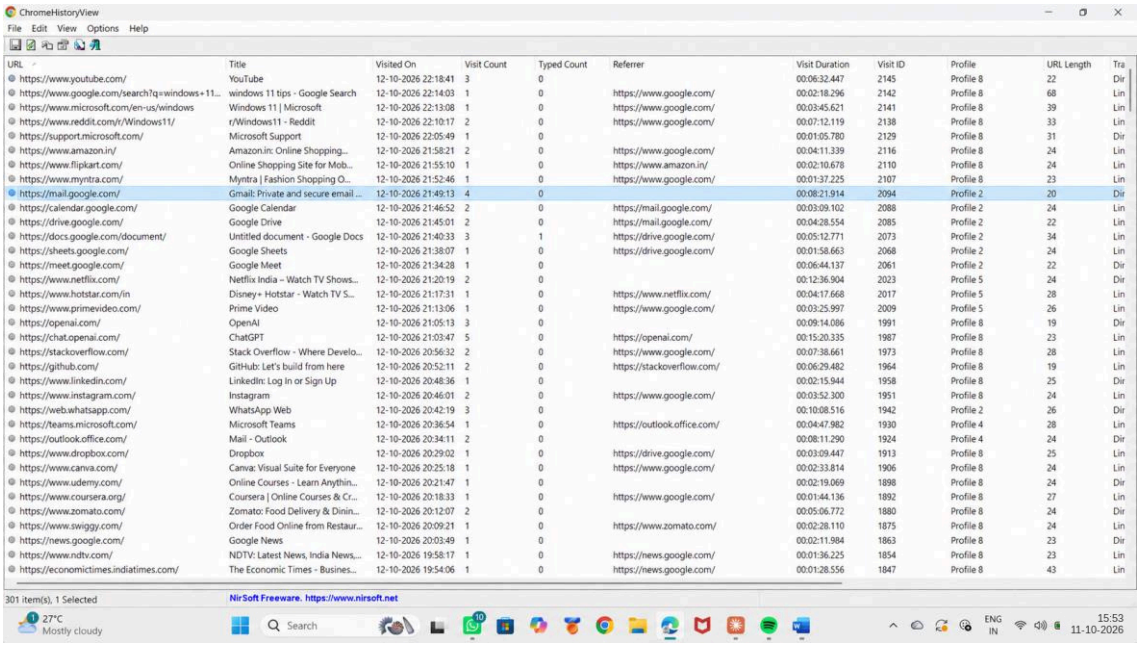


Fig. 8.1 ChromeHistoryView – list of visited web pages

TIMESTAMP: 11-10-2026, 15:53

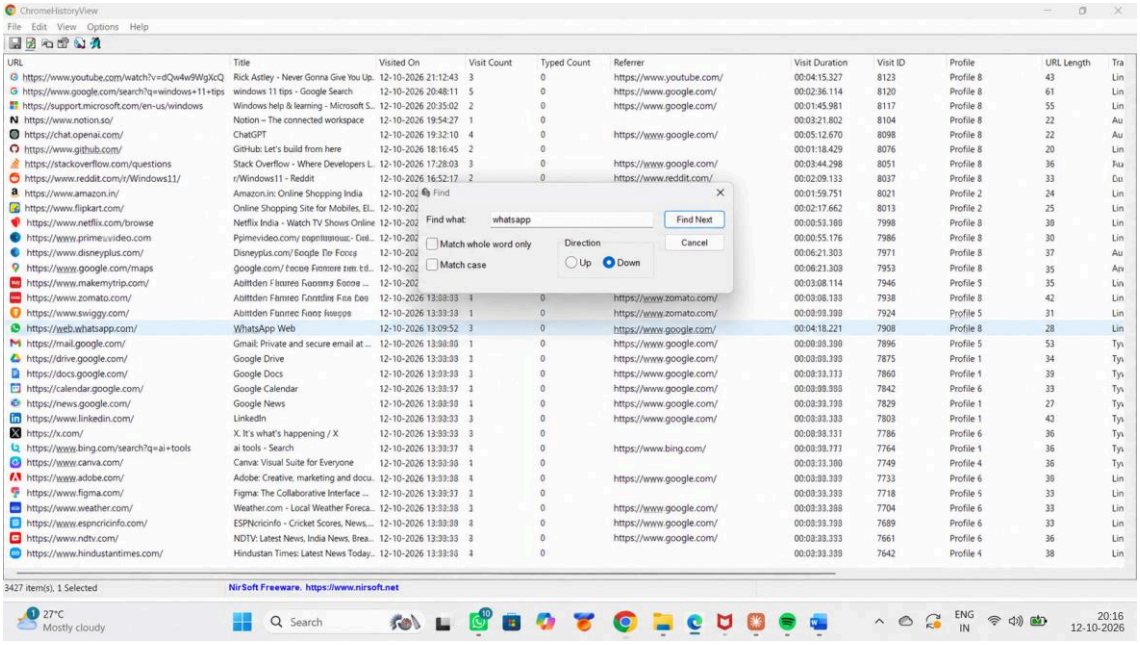


Fig. 8.2 ChromeHistoryView – searching the history for the keyword ‘whatsapp’

TIMESTAMP: 12-10-2026, 20:16

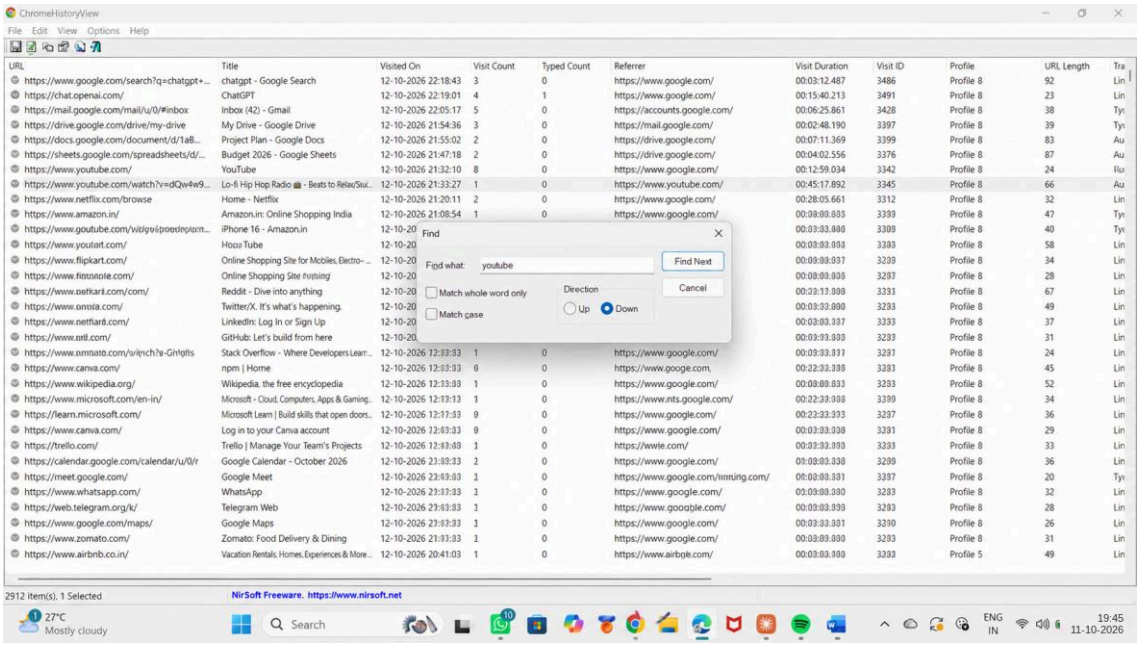


Fig. 8.3 ChromeHistoryView – searching the history for the keyword 'youtube'
TIMESTAMP: 11-10-2026, 19:45

Result

Hence, the browser artifacts were extracted successfully. The visited web pages, along with their visit time, visit count and referrer, were retrieved and can serve as evidence during a forensic investigation.

EXPERIMENT – 9

Comparison of Two Files by Compare It Software

Aim

To compare two files for forensic purposes using the Compare It software and to produce a report of the differences between them.

Procedure

- Install the Compare It software on the system.
- Compare It places two files side by side and highlights the differing sections in colour, which makes the analysis easier.
- Open the Compare It tool; a dialog box appears for choosing the files to be compared.
- Select the first file and click ‘Open’, then select the second file and click ‘Open’.
- The changes between the two files are marked with highlighted bars, and any change can be moved from one file to the other with a single mouse click or keystroke.
- The files can also be edited directly within the comparison window.
- Print or save a colour report of the differences exactly as they appear on the screen.

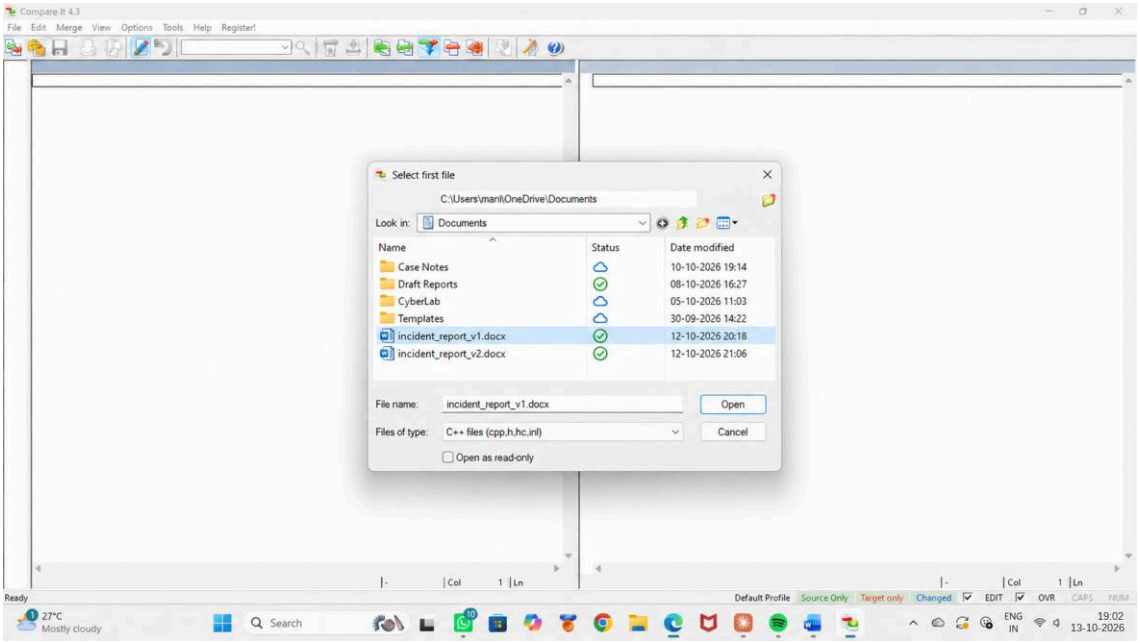


Fig. 9.1 Compare It – choosing the first file for comparison

TIMESTAMP: 13-10-2026, 19:02

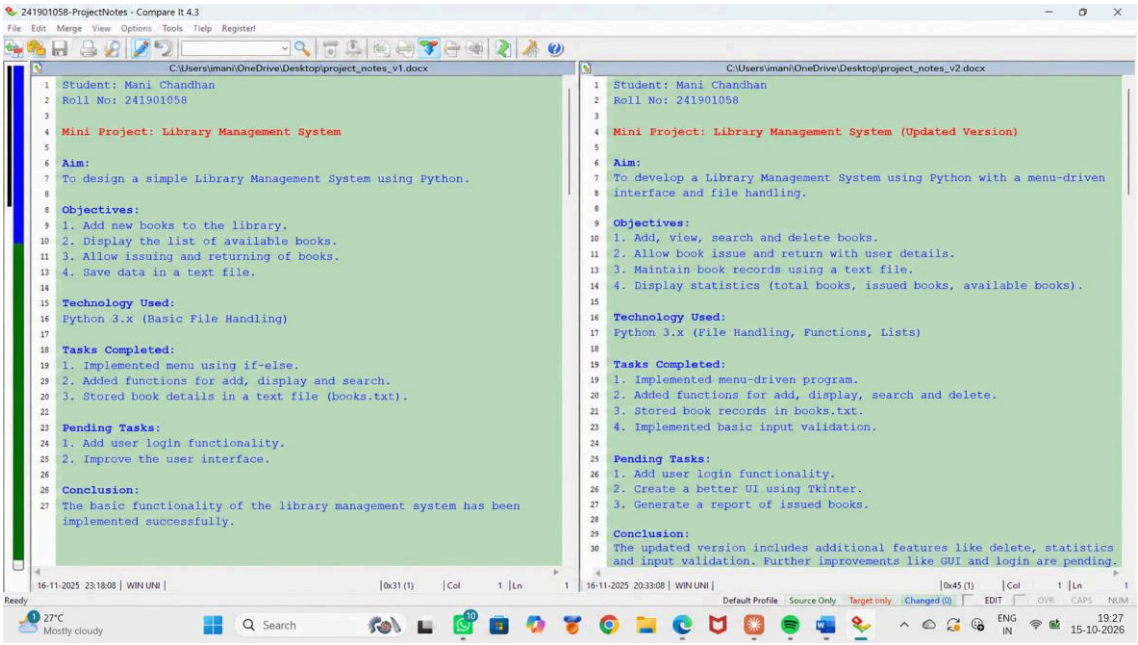


Fig. 9.2 Compare It – side-by-side comparison highlighting the differences between the two files

TIMESTAMP: 15-10-2026, 19:27

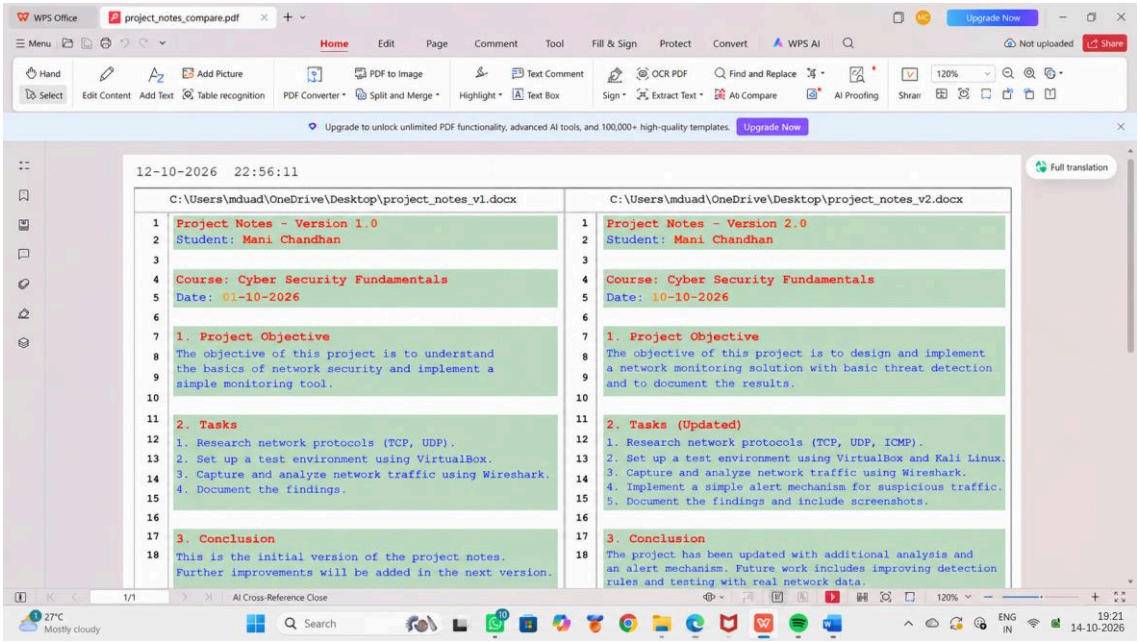


Fig. 9.3 Generated comparison report (project_notes_compare.pdf)

TIMESTAMP: 14-10-2026, 19:21

Result

Hence, the two files were compared successfully with Compare It. The differences between them were highlighted and a comparison report was produced for forensic investigation.

EXPERIMENT – 10

Collecting Email Evidence in Victim PC

Aim

To gather e-mail evidence from a victim's computer by acquiring its physical memory and extracting the e-mail artifacts from it.

Procedure

- **Memory acquisition:** The first step is to capture the RAM of the victim system using the DumpIt utility.
- DumpIt produces a physical memory dump of a Windows machine, runs on both 32-bit (x86) and 64-bit (x64) systems, and is convenient to carry on a USB drive for quick incident response.
- Run DumpIt.exe. It displays the address space size, the free space and the destination path of the raw dump, and asks for confirmation before continuing.
- Type 'y' and press Enter; the memory dump (.raw) is created and saved in the same folder.
- **Feature extraction:** Download and install Bulk Extractor Viewer from GitHub.
- Launch Bulk Extractor Viewer and choose to generate a report; give the DumpIt memory image as the input, select an output folder and click 'Start'.
- Bulk Extractor scans the image and stores its findings in separate feature files such as email.txt, email_histogram.txt, url.txt and telephone.txt.
- Open email.txt to examine the e-mail addresses found on the victim system, and email_histogram.txt to see how often each address appears.

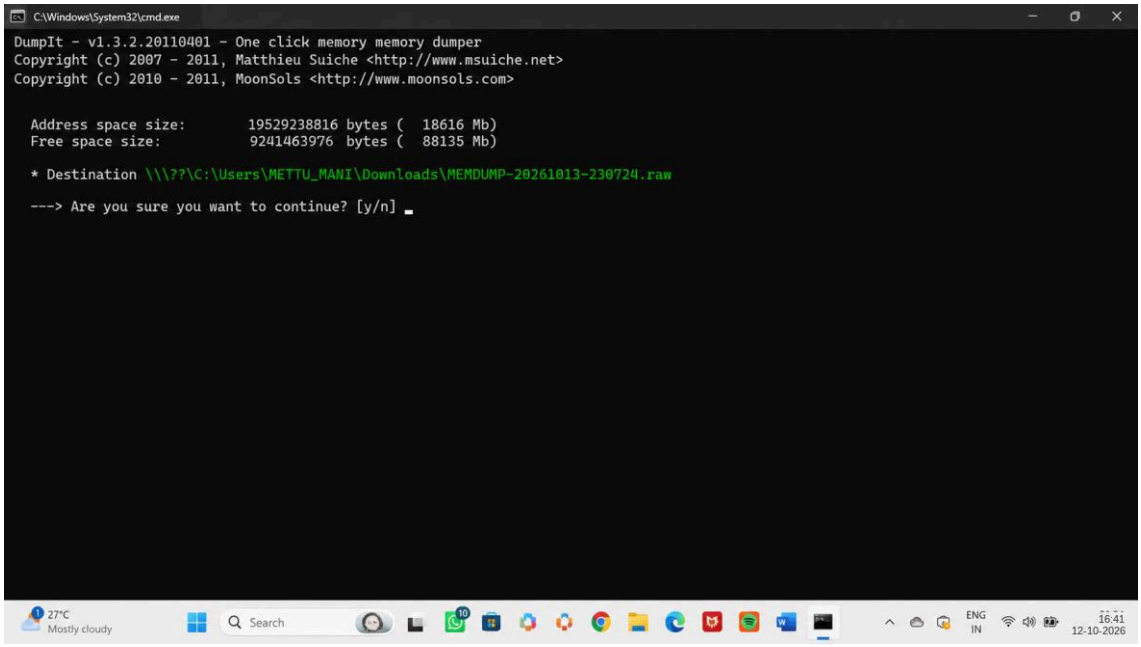


Fig. 10.1 DumpIt – destination of the raw memory dump and the confirmation prompt
TIMESTAMP: 12-10-2026, 16:41

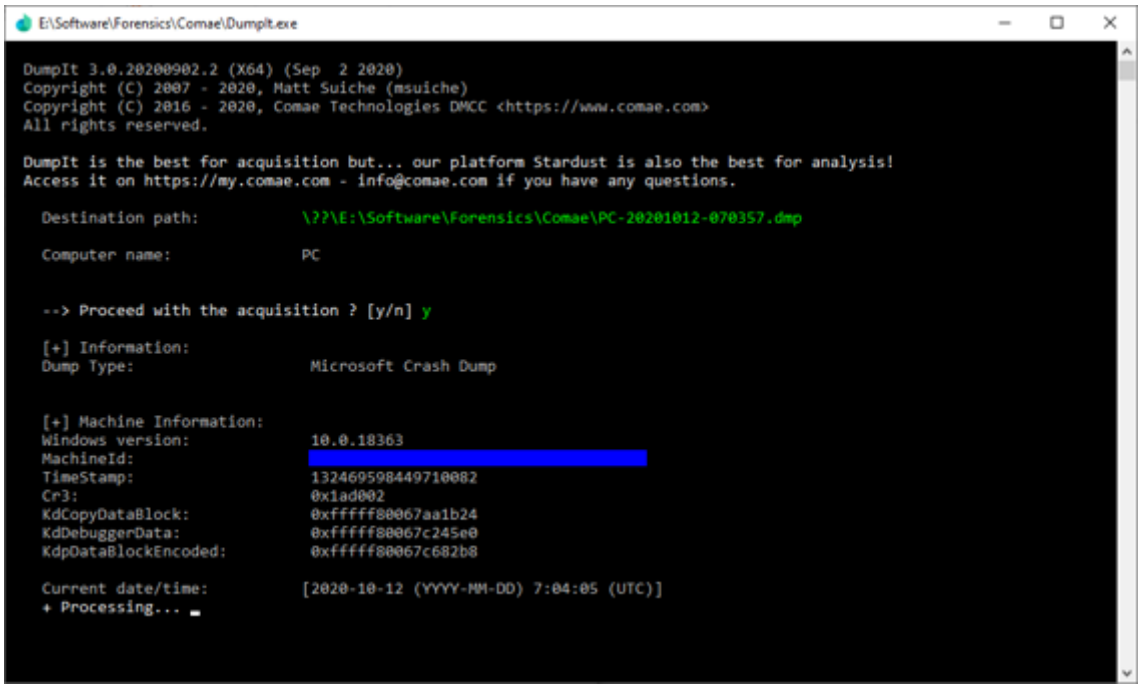


Fig. 10.2 DumpIt – physical memory acquisition in progress



Fig. 10.3 Bulk Extractor Viewer – main window used to open the generated reports

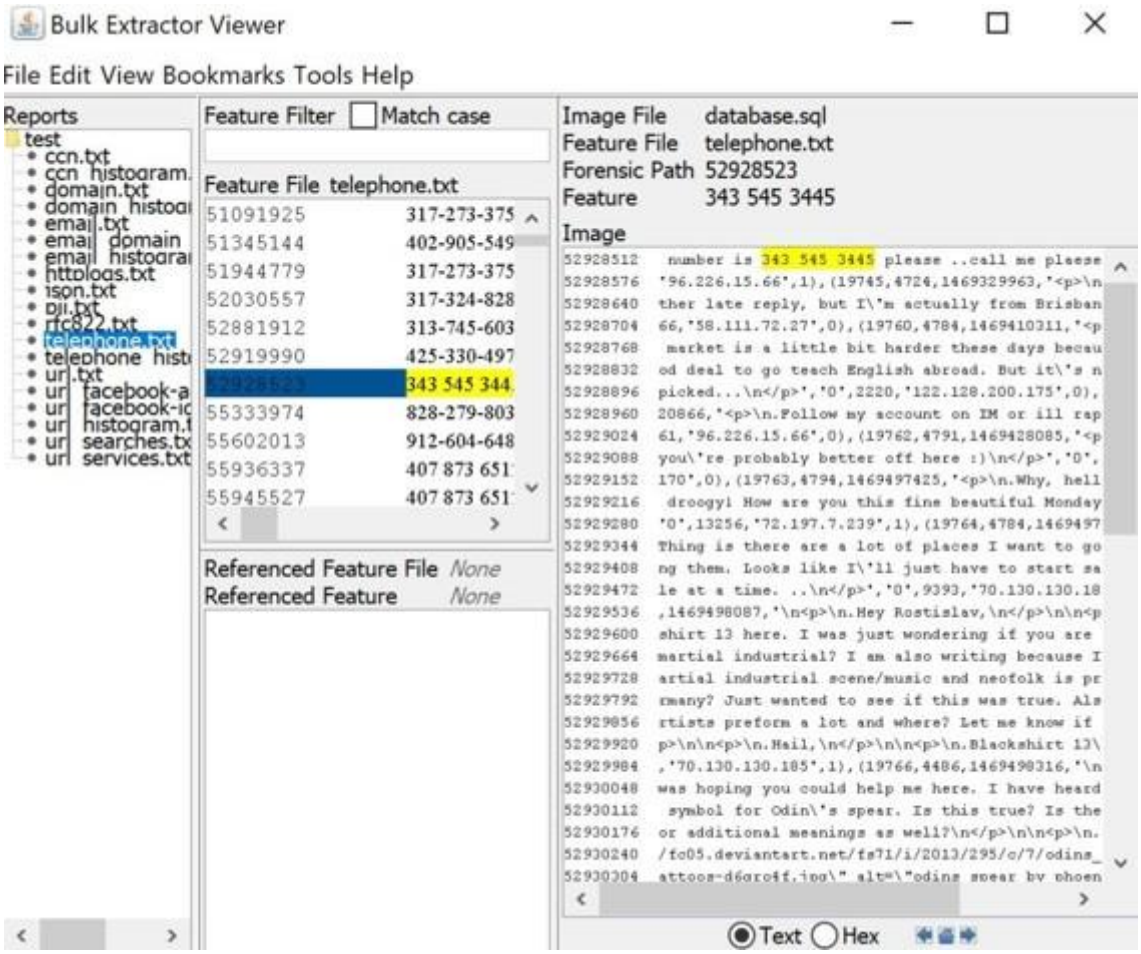


Fig. 10.4 Bulk Extractor Viewer – viewing the extracted feature files of the report

Result

Hence, the e-mail evidence was collected from the victim's PC. The physical memory of the system was acquired with DumpIt, the memory image was processed with Bulk Extractor, and the extracted e-mail addresses and their histogram were examined.

EXPERIMENT – 11

Hiding and Extracting a Text File behind an Image / Audio File using Command Prompt

Aim

To learn how a text file can be concealed behind an image file or an audio file and later extracted, using the Windows Command Prompt.

Procedure

- **Hiding a text file in an image:** Windows allows any file (.txt, .zip, .rar, .jpg, etc.) to be joined to another file through a binary copy, so a text file can be concealed inside an image directly from the Command Prompt.
- Place the file to be hidden (alert.txt) and the carrier image (coastline.jpg) in the same folder (Case_Files).
- Open the Command Prompt (Win + R → cmd) and move to that folder using the cd command.
- Join the two files with the binary copy command: copy /b alert.txt + coastline.jpg hidden_note.jpg
- The output file hidden_note.jpg is created; it still opens as an ordinary image while carrying the text data inside it.
- **Extracting the hidden data:** Right-click hidden_note.jpg and open it with Notepad; the raw bytes of the combined file are displayed and the concealed text can be located within them.
- **Packaging the evidence:** A case manifest is written using echo commands (for example echo Case: MANII-2026-10-14 > case_manifest.txt), and all the files are compressed into a single archive with: tar -a -c -f evidence_pack.zip alert.txt coastline.jpg hidden_note.jpg case_manifest.txt
- The contents of the archive are listed with tar -tf evidence_pack.zip, and the folder is checked with dir /b to confirm that every file is present.
- **Hiding data in an audio file:** The same technique works with an audio carrier, for example copy /b sound.mp3 + secret.zip newfile.mp3. The hidden archive is recovered by renaming the file to .zip or by opening it with an archive program.
- On Linux the same merge can be done with: cat sound.mp3 secret.zip > newfile.mp3

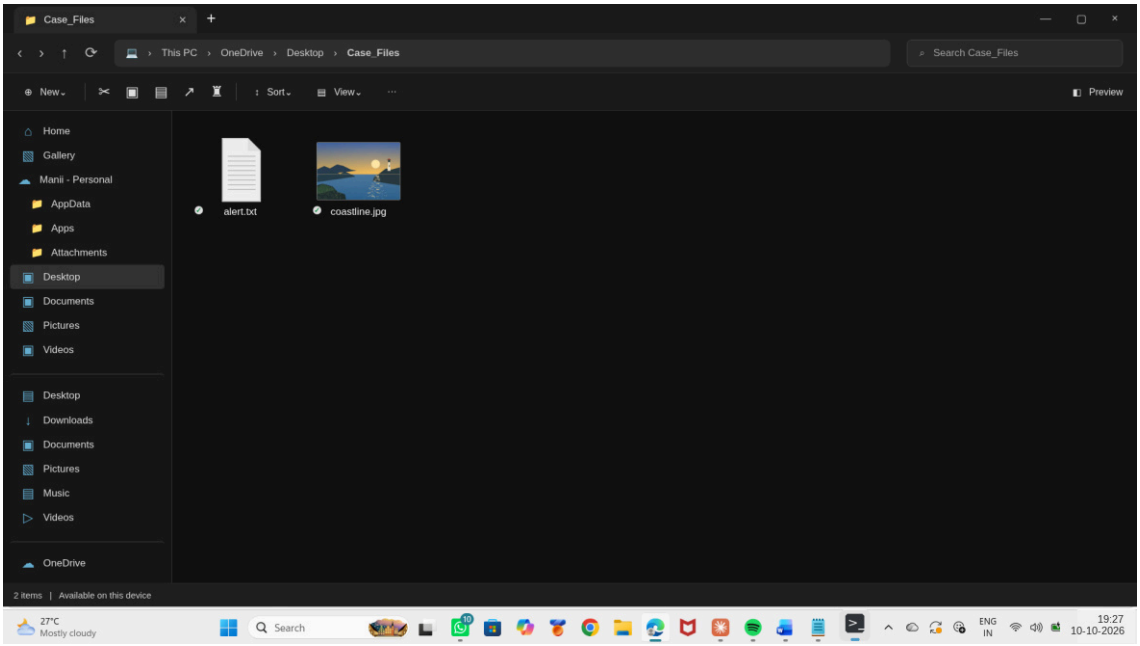


Fig. 11.1 The text file alert.txt and the carrier image coastline.jpg in the Case_Files folder

TIMESTAMP: 10-10-2026, 19:27

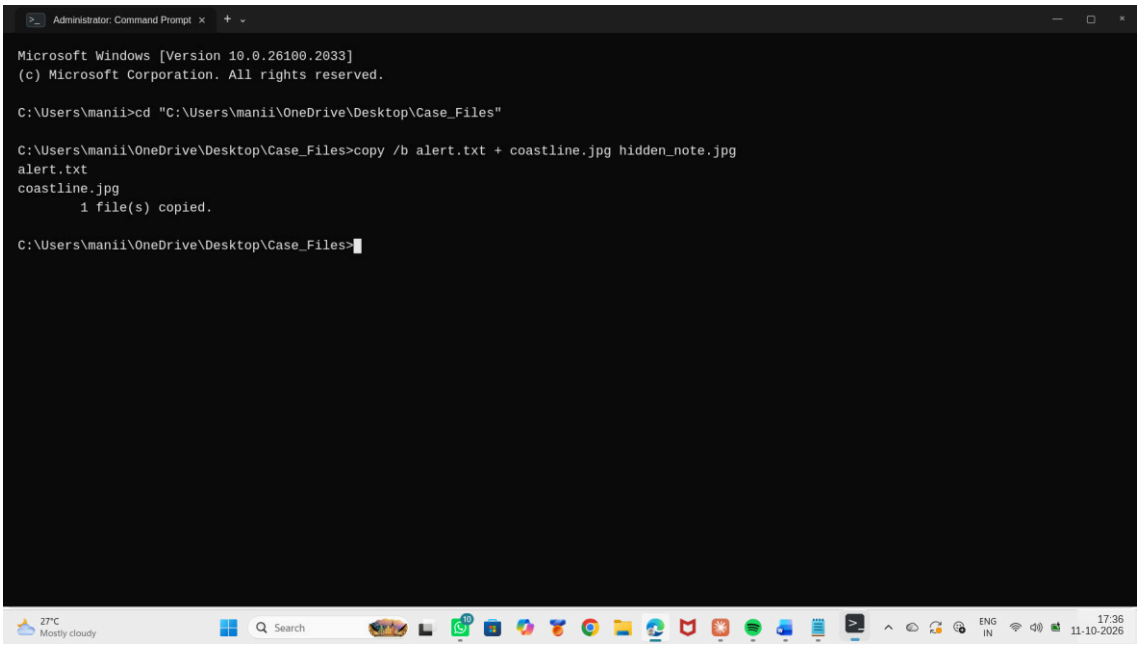


Fig. 11.2 Command Prompt – merging the text file with the image using copy /b

TIMESTAMP: 11-10-2026, 17:36

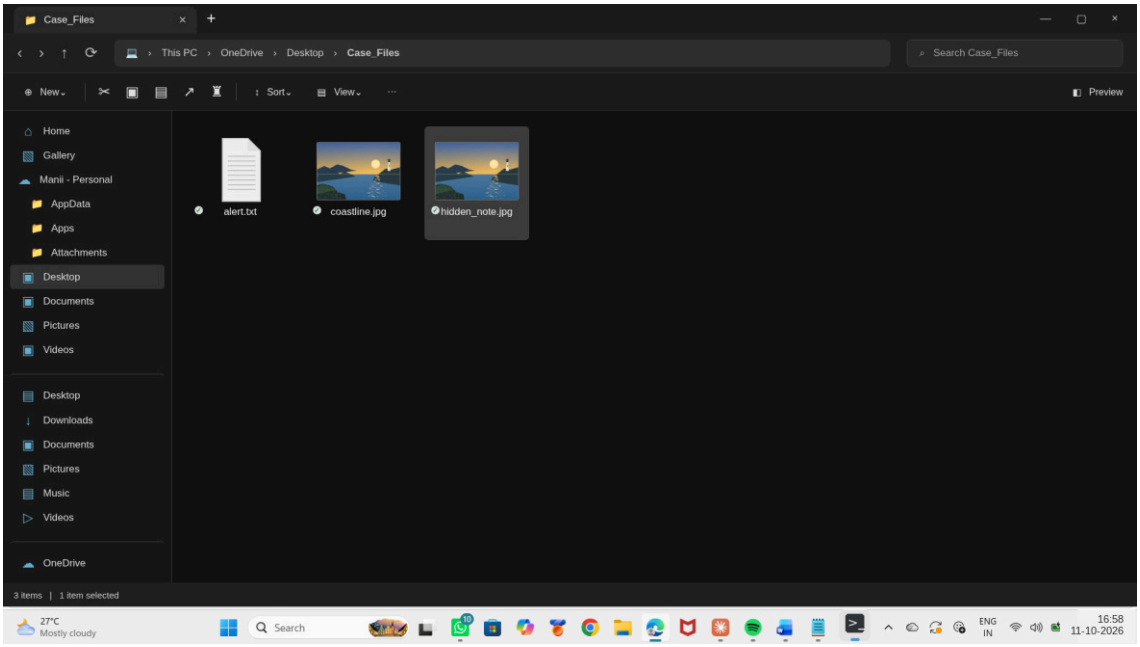


Fig. 11.3 The output image hidden_note.jpg created after the merge

TIMESTAMP: 11-10-2026, 16:58

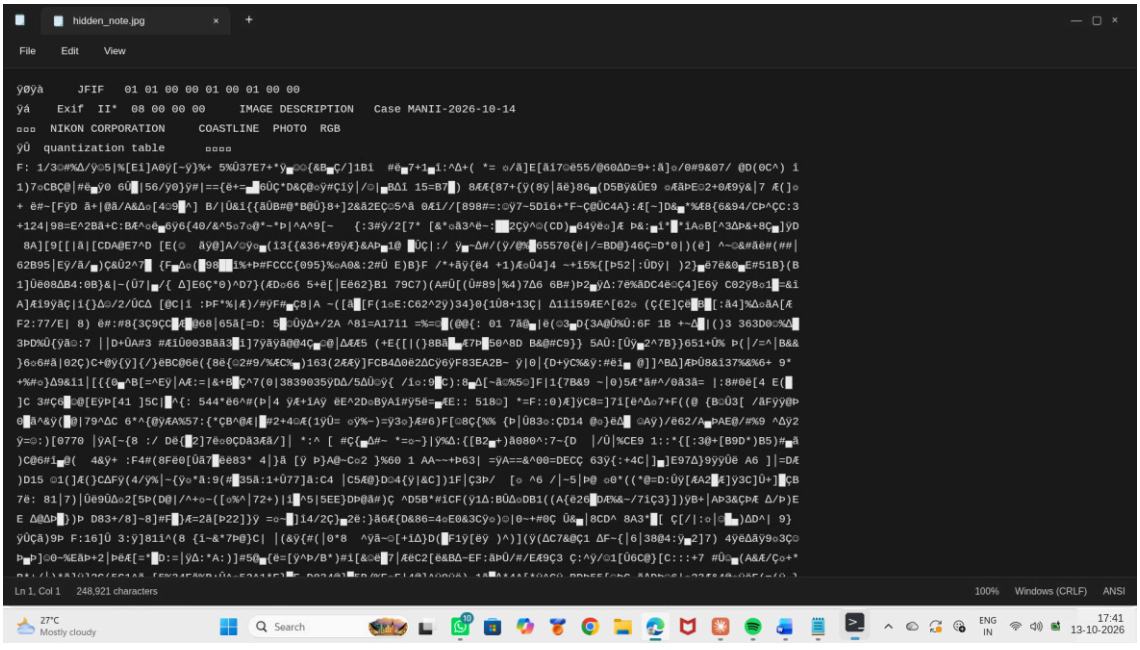


Fig. 11.4 Opening hidden_note.jpg with Notepad to inspect the hidden content

TIMESTAMP: 13-10-2026, 17:41

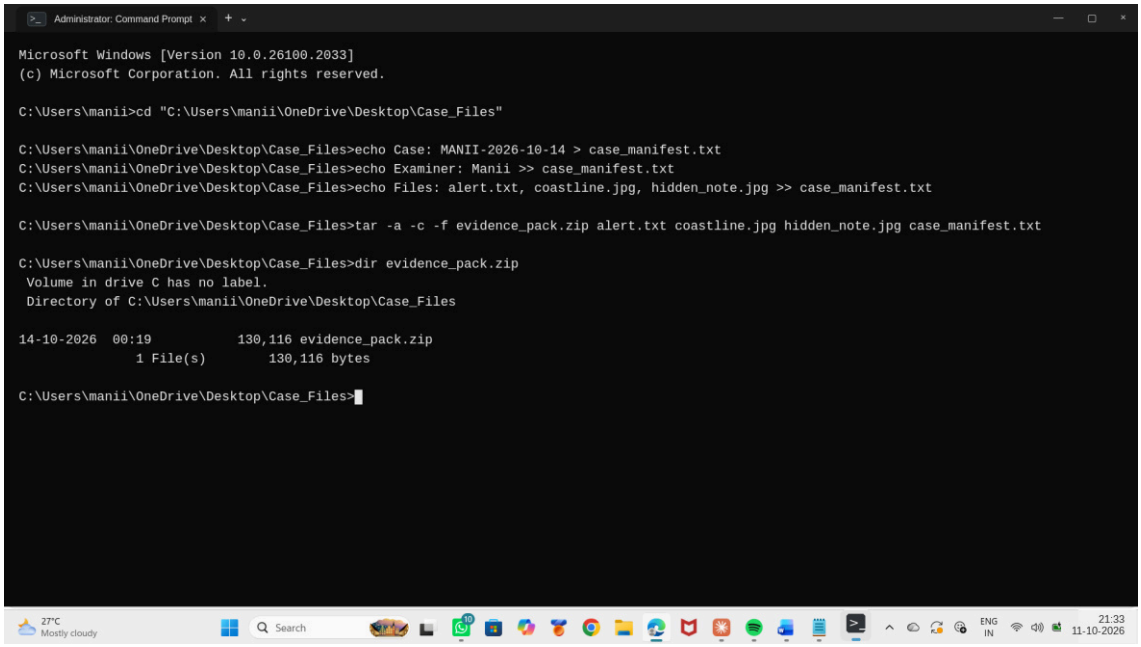


Fig. 11.5 Command Prompt – creating the case manifest and packaging the files into evidence_pack.zip

TIMESTAMP: 11-10-2026, 21:33

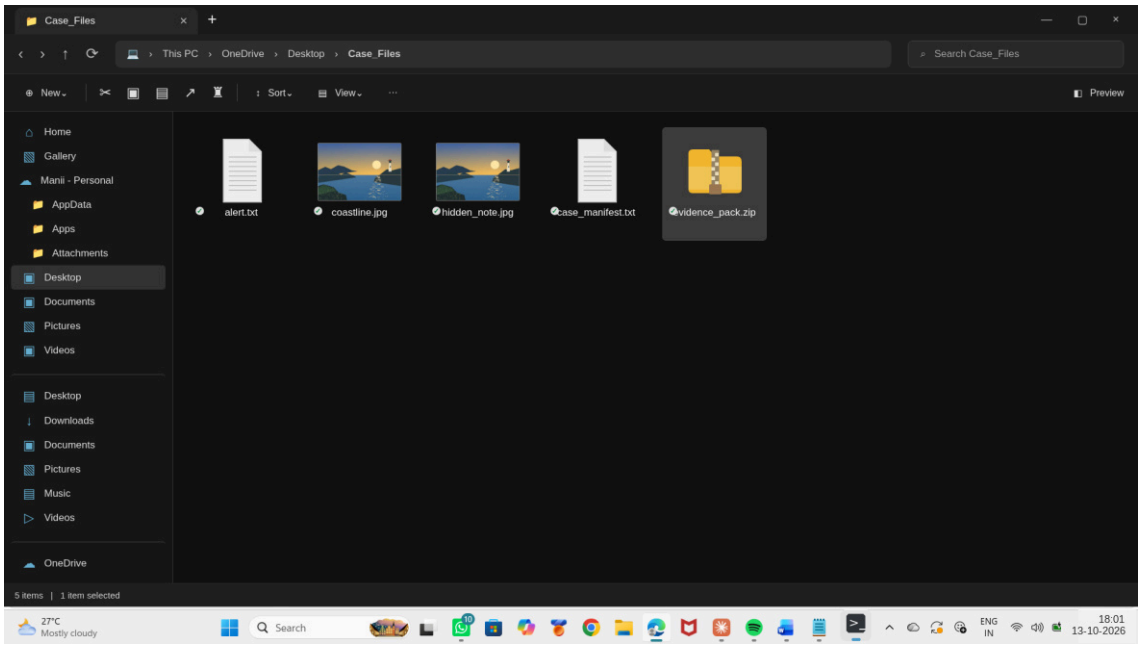


Fig. 11.6 The Case_Files folder showing the manifest and the evidence archive

TIMESTAMP: 13-10-2026, 18:01

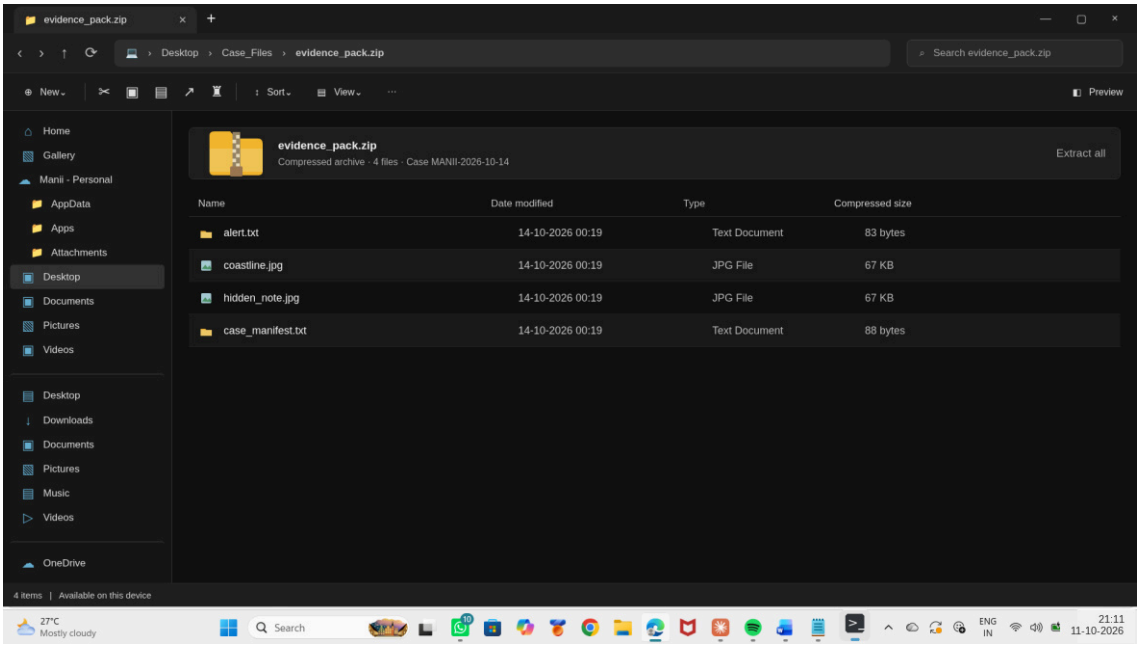


Fig. 11.7 Contents of evidence_pack.zip viewed in File Explorer

TIMESTAMP: 11-10-2026, 21:11

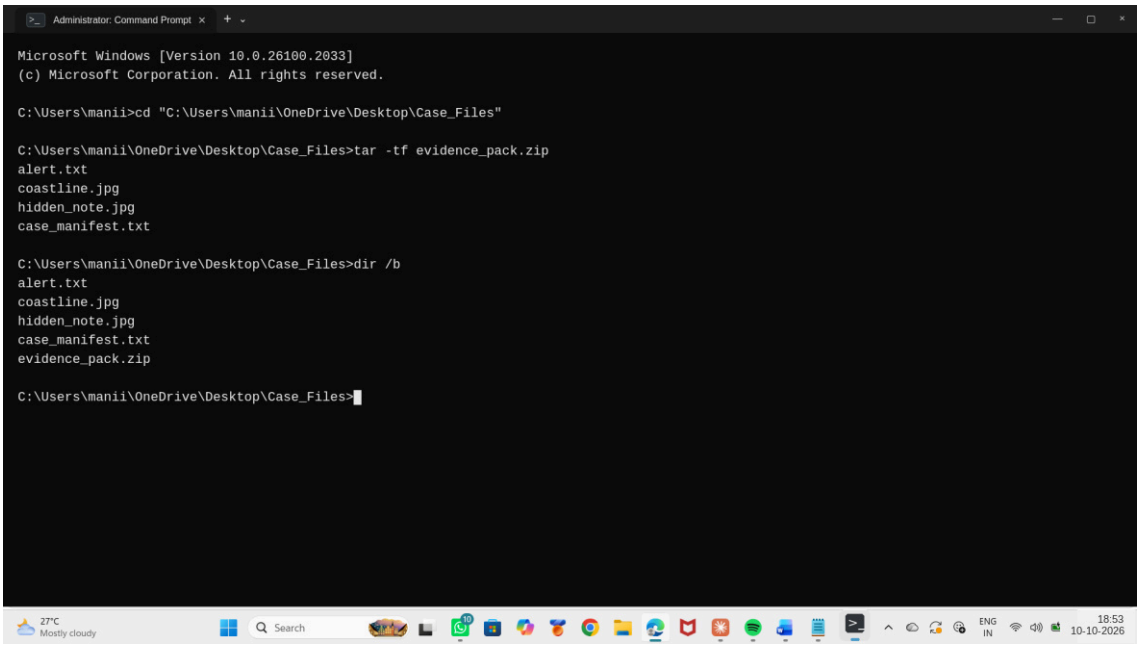


Fig. 11.8 Command Prompt – listing the archive contents with tar -tf and dir /b

TIMESTAMP: 10-10-2026, 18:53

Result

Hence, a text file was concealed behind an image file using the Command Prompt and the hidden content was retrieved, while the merged image continued to open as a normal picture. The evidence files were also packaged and verified in a ZIP archive, and the same technique was understood for audio carriers.

EXPERIMENT – 12

Extraction of EXIF Data from Image Files using ExifReader

Aim

To retrieve the Exchangeable Image File Format (EXIF) data stored in an image file using the ExifReader software.

Introduction

When a computer, phone or other digital device is seized as evidence, it frequently holds photographs that may be relevant to the case. Most cameras and phones embed descriptive information into every image they capture, which can reveal the who, what, when and where of the photograph. This embedded information, known as EXIF data, is often of great value to a forensic examiner.

EXIF is a standard defined by the digital camera industry for storing metadata in image and sound files. It records details such as the camera settings, date and time, shutter speed, exposure, flash usage, compression, the camera make and model, and more. Several programs can read this data; in this experiment a simple Windows utility named ExifReader is used.

Procedure

- Download ExifReader and run ExifRead.exe; a clean and simple interface opens.
- Click ‘Open’ and browse to the image stored on the system or on the seized media. JPEG/JPG files usually carry the richest metadata.
- Once the image is loaded, its thumbnail appears on the left and the EXIF fields are listed on the right.
- Examine the extracted fields such as the camera make and model, date and time of capture (DateTime and DateTimeOriginal), exposure time, F-number, exposure program, ISO speed, compressed bits per pixel, focal length, flash and the GPS location (when present).
- Record the details that are relevant to the investigation, since EXIF data can show when, where and with which device a photograph was taken.

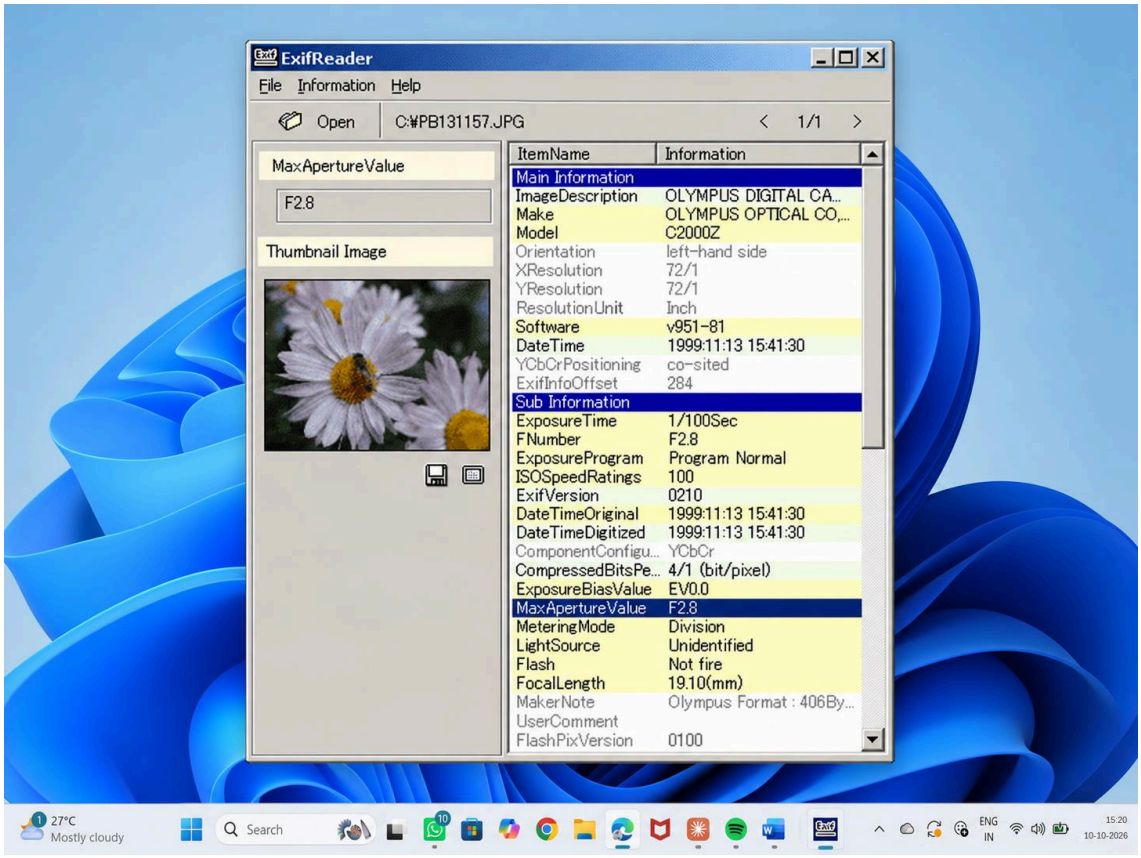


Fig. 12.1 ExifReader – EXIF metadata extracted from a JPEG image

TIMESTAMP: 10-10-2026, 15:20

Result

Hence, the EXIF data of the image file was extracted successfully using ExifReader. The camera make and model, the date and time of capture and the photographic settings were obtained, which helps the investigator link an image to its source device.